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Multifaceted structural classification of Indian legal judgments

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Abstract
Automatically identifying structural components of Indian court judgments is critical for effective legal document analysis but remains challenging due to complex legal language, class imbalance, and limited annotated data. This paper proposes a multiclass structural classification framework for Indian legal judgments using a dataset of nearly 6500 judgment segments from Indian Kanoon, manually annotated into 15 structural categories. A domain-specific lexicon of 5,000 legal n-grams is used to support feature construction for TF-IDF representation. We evaluate statistical representations (TF-IDF), dimensionality reduction (PCA), data augmentation, and contextual embeddings from transformer-based models across multiple machine learning and deep learning classifiers. The best-performing models are further combined using interpolation-based fusion. Experimental results show that a fused Legal-BERT and Indian Legal-BERT model achieves the best performance, with 84% accuracy, 84% weighted F1-score, and 80% macro recall, without data augmentation or manual feature engineering. Performance gains are validated using a paired Wilcoxon signed-rank test ($p < 0.05$), demonstrating robust and consistent improvements across structural classes. Further, explainability tools are used for interpreting and understanding the primary tokens influencing the model's decisions.

Keywords Structural analysis, Court judgments, Machine Learning, Deep Learning, Word embeddings, N-gram Term Frequency-Inverse Document Frequency, Transformer Models

1 Introduction

According to the National Judicial Data Grid (NJDG), Indian judicial system is currently under considerable pressure due to an unprecedented accumulation of pending cases, with more than 44 million matters awaiting resolution [1]. The growing backlog of court cases places heavy pressure on the judicial system and increases the workload of legal professionals. It is time-consuming and challenging for legal professionals to read long and complex court judgments to find information relevant to their cases. The large number of cases and the complexity of legal procedures further delay the delivery of legal services in India. Artificial intelligence (AI) can play a supportive role by automating legal

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document analysis, classification, and summarization [2, 3]. AI-based systems are able to process large collections of legal text within a relatively short time, identify information of interest, and reduce the effort required for detailed legal reasoning. In practice, legal professionals can work away from manual document review and focus more on case preparation and interaction with clients. However, the automatic analysis of legal judgments remains challenging due to several factors. Legal documents are typically long, rely on specialized legal terminology, and depend heavily on context that spans multiple sections. Therefore, effective legal text mining methods and context-aware models are necessary to support reliable legal judgment analysis [4].

The digitization of legal records has made legal information easier to access and manage in everyday practice. This development has strengthened the connection between legal document analysis and technological tools, particularly in terms of supporting informed decision-making and improving access to justice [5]. Within this setting, the structural classification of court judgments has shown value for legal research, judicial reasoning, legal education, and public access to legal materials [6].

Legal judgment structuring refers to the manner in which courts arrange and present reasoning, evidence, and conclusions that are internally coherent. When judgments clearly separate elements such as case facts, legal arguments, judicial reasoning, precedents, and final rulings, they tend to be easier to follow and interpret. Breaking a judgment into identifiable sections such as issues, arguments, precedent analysis, court discourse, and judicial opinions can help legal practitioners locate relevant portions of the text with less effort. This in turn, supports more timely and accurate decision-making [7]. The usefulness of structured judgments is not limited to professional practice. They additionally assist with legal education, make the legal system more open to the public, and support more extensive efforts to reform the courts while giving legal analytics a more stable foundation. From a computational standpoint, an organized document structure simplifies automated processing and enables the incorporation of legal documents into broader analytical workflows, enhancing the efficiency and accessibility of legal systems [8, 9].

Recent advancements in machine learning, deep learning [10, 11] and transformer-based architectures [12] have transformed the automatic analysis of legal documents [13]. These models represent complex linguistic patterns and handle dependencies that extend across long variants of text, which is a common characteristic of legal judgments and statutes [14]. As a result, ML and DL approaches are used more frequently in practice to support tasks such as document review, legal research, information extraction, and the analysis of case outcomes over large legal text collections.

Beyond task automation, learning-based models are used for structural analysis of legal documents and represent relationships among legal concepts, entities, and argumentative elements. Such representations are important for interpretative tasks that rely on consistency and contextual alignment throughout a document [15]. Artificial intelligence advancements have led to legal systems that are efficient and transparent [16]. Legal information is nowadays more accessible for both lawyers and the general public. Lawyers can get useful information from vast legal datasets using predictive analytics and data-driven mining methodologies. This facilitates evidence-based decision-making, risk evaluation, and strategic planning in litigation scenarios [17]. This work builds on existing developments by focusing on a detailed structural classification of Indian law

judgments. The novelty and research contributions of the proposed approach are outlined in the following section.

1.1 Novelty and research contributions

Court judgments issued in India are often lengthy and lack a consistent organizational pattern. Details concerning the case background, legal issues, judicial reasoning, and final decisions are commonly spread across the document rather than confined to clearly marked sections. This makes the automatic assignment of text segments to predefined structural categories a difficult problem in practice. The challenge is further aggravated by the use of specialized legal terminology, contextual dependencies that span multiple sentences or paragraphs, pronounced imbalance among different structural sections, and the limited availability of carefully annotated Indian legal datasets. Together, these conditions necessitate sequence-aware classification approaches that can account for both local textual cues and wider contextual information in order to support accurate and scalable analysis of Indian legal judgments.

This study introduces a single framework aimed at fine-grained structural classification of Indian legal judgments, with the text divided into 15 predefined categories. The framework combines statistical text representations with semantic and contextual features and evaluates both conventional ML methods and more recent DL architectures. The dataset is sourced from Indian Kanoon and includes a varied set of court judgments, along with a domain-specific list of roughly 5,000 commonly occurring legal n-gram expressions. Empirical evaluation is carried out on close to 6,500 judgment segments that were manually annotated by legal experts. While earlier work, such as [18], established baseline approaches for structuring Indian legal judgments, the present study extends this line of research through refinements in methodology and a broader empirical assessment. The following section describes the main contributions and methodological advances in detail.

- A comprehensive analysis is conducted using statistical word representations based on n-gram-driven TF-IDF features extracted from frequently occurring legal phrases. The study further investigates the impact of data augmentation and dimensionality reduction techniques, followed by classification using machine learning models.
- Semantic representations are learned through contextual embedding derived from BERT-based models, including RoBERTa, DistilBERT, BERT, Legal-BERT, and Indian Legal-BERT. These embeddings are subsequently evaluated using both ML and DL classifiers
- Multiple advanced BERT-based language models are fine-tuned specifically for the task of structural judgment classification, with detailed analysis performed at class level to understand their behavior across structural categories.
- The best-performing fine-tuned BERT models are further improved through model ensembling strategies based on linear interpolation techniques such as SLERP, TIES, and DARE-TIES.
- Extensive class-wise evaluation of the top-performing models is performed using detailed classification metrics, validation curves, and statistical significance analysis, followed by an in-depth discussion of performance trends across different structural classes.

- A post-hoc explainability study is conducted on the top performing model using explainable AI tools viz. SHapley Additive exPlanations (SHAP) [19] and Integrated Gradients (IG) [20].

This paper examines the application of artificial intelligence methods to the structural classification of Indian legal judgments. Related work in legal analytics, with an emphasis on AI-based techniques, is discussed in Sect. 2. The dataset used in this study is introduced in Sect. 3. Section 4 describes the text representation methods employed such as n-gram based TF-IDF features and contextual word embeddings along with the construction of the proposed models. Results from the various model configurations are presented in Sect. 5 and discussed in Sect. 6, where differences in performance are discussed. Section 7 concludes the paper by summarizing the main findings and indicating possible directions for future work.

2 Related work

The use of artificial intelligence in the legal domain represents a gradual but meaningful shift in legal analytics [21]. Initial work in this area focused mainly on converting legal records into digital form. Over time, however, attention has shifted towards AI techniques that attempt to engage with the semantic content of legal texts rather than relying solely on surface-level representations. Today, AI systems are applied in a variety of legal settings, ranging from the automation of routine, repetitive tasks to providing analytical support for predictive and decision-oriented processes in more complex legal contexts [22].

2.1 AI-based legal analysis

Early uses of AI in legal applications were mostly confined to straightforward keyword-based document searches. As ML methods [23] and natural language processing techniques matured [24], these systems gradually moved beyond simple lexical matching and began to incorporate contextual cues and more nuanced linguistic structures present in legal text [25, 26]. More recently, AI has been extended to tasks that place higher demands on modeling capability, such as case outcome prediction, risk analysis, and support for legal research. This shift has been driven by progress in DL models [22], neural network based methods [27], and advanced language modeling approaches. Through the automation of repetitive operations and the identification of patterns across large legal corpora, these techniques contribute to more efficient workflows and enable data informed decision making for legal practitioners [24].

Recent work suggests a gradual improvement in the capacity of artificial intelligence systems to model both the meaning and structural organization of legal documents. Techniques such as sentiment analysis [28], named entity recognition, legal judgment summarization [29], and relation extraction have been applied across several tasks, including legal research, case summarization, and the analysis of precedents [3]. A recurring goal within legal analytics is the transformation of largely unstructured legal text into structured representations that can be more readily used for analysis and decision support [16].

Initial studies in this area made use of techniques such as distinct expression identification [30] and conventional ML models, including support vector machines [31]. Over time, research has shifted towards sequence-to-sequence architectures aimed at

extracting legal entities and role information directly from text [8, 32]. Further improvements have been observed when semantic analysis is combined with sentiment-focused methods [33, 34], as well as through the adoption of knowledge-driven and concept-based approaches for fact extraction and legal document retrieval [21, 35, 36]. More recent frameworks, including joint entity relation extraction models [7] and systems such as ALEM [37], have reported stronger performance than earlier techniques, supporting the development of more reliable legal analytics systems [38].

Document classification occupies a central role in legal analytics [39], largely because of its relevance to interpreting legal texts and supporting judicial outcome prediction. Early studies [17, 40] highlighted the feasibility of applying computational techniques to legal analysis. As issues such as synonymy and variation in legal terminology became more apparent, semantic web based frameworks, including LegalRuleML [41], were introduced to address these limitations. At the same time, progress in NLP and conceptual modeling has contributed to the development of classification approaches that emphasize interpretability and adaptability [42].

Ontology-based methods have been used to represent and summarize structured knowledge present in legal judgments, supporting more interpretable classification results [43, 44]. Subsequent studies suggest that the inclusion of text summarization can improve judgment prediction performance [45]. In parallel, semantic representations derived from ML and DL models have shown effectiveness in related tasks, such as law article prediction [46]. Beyond the analysis of judgments themselves, these approaches have been applied across a broader set of legal applications, including patent analytics [47], automated legal document classification [48], case outcome prediction, sentiment analysis of judicial texts [49], and the organization of large-scale legal document collections.

2.2 Deep learning approaches

During the past decade, DL architectures such as Recurrent Neural Networks (RNNs), Gated Recurrent Units (GRUs), Long Short-Term Memory (LSTM) networks, and Convolutional Neural Networks (CNNs) have been investigated for legal document classification [50]. Several studies have applied these models to the structural classification of civil and criminal judgments [51]. Survey based analyses further suggest a growing level of adoption of these approaches within the Indian legal research landscape [52].

More recent studies have combined BERT-based representations with LSTM and CNN architectures for legal judgment prediction, with reported accuracies reaching up to 75% [53]. Beyond these hybrid configurations, other approaches such as PRTIRCD [17] and graph-enhanced LSTM models that incorporate domain knowledge, have shown additional performance improvements, particularly in Chinese legal documents [40]. Alongside judgment classification, DL techniques have also been applied to related legal tasks, including information retrieval, document analytics, and decision support systems [54–56], as well as sentiment analysis using bidirectional LSTM and GRU based models.

Research beyond the Indian legal context has addressed similar problems in other jurisdictions. In Germany, a hybrid approach combining rule-based methods with machine learning was proposed to classify civil law norms into nine semantic categories, with an explicit emphasis on interpretability achieved through local linear

approximation techniques [57]. In the Canadian legal domain, recent work [58, 59] has examined deep learning based sentiment analysis for maritime case law, using CNN and LSTM models to promote more consistent judicial decision-making and to reduce workload. One such study reported a classification accuracy of 98.05%, alongside observed gains in legal case processing efficiency [13].

2.3 Advanced large language model-based studies

In addition to traditional DL and ML models, recent advancements have seen the use of DL in computer vision applications for legal document review. This includes image classification, clustering, and object detection using transfer learning techniques [60]. Furthermore, DL mixed with word embeddings has become more effective in reducing human labeling efforts and manual review of documents [61, 62]. Recent research efforts have been directed towards utilizing large language models (LLMs) for unstructured legal document classification [63]. The study in [63] introduces MESc, a deep learning-based hierarchical framework for predicting legal judgments from long, unstructured case documents. By using embeddings from large language models like GPT-Neo and GPT-J, combined with unsupervised clustering and transformer encoders, the method outperforms previous approaches, achieving a performance improvement of at least 2 points across various legal datasets from India, the EU, and the US. Transformer-based embeddings combined with classification have been increasingly used in legal case outcome prediction [64]. The researchers in [65] explore the use of abstractive summarization models, including LLMs, for summarizing complex legal case judgments from the UK and Indian Supreme Courts, as well as US government reports. While LLMs and abstractive models generally outperform extractive methods, they still produce inconsistencies and hallucinations, suggesting that further improvements are needed, with a human-in-the-loop approach being more reliable for ensuring summary accuracy.

An analysis of various LLMs is conducted for structural classification of Indian legal judgments using the Indian Kanoon dataset in [18]. An extractive summarization of Indian legal judgments using generative AI and NLP methods is proposed in [66, 67]. The results indicate that the OpenHermes–2.5-Mistral-7B-16k-GGUF model was the top performer with an accuracy of 61% with zero-shot prompting. This is the first attempt in literature towards structural classification of Indian legal judgments. The LM-EGAC method, which combines optimized topic modeling and ensemble gradient boosting, has been proposed for classifying legal documents, showing better precision and reduced error compared to traditional methods [68]. Transformer models, such as BERT, RoBERTa, and XLNet, have been widely adopted for legal judgment prediction due to their ability to handle complex legal texts. Recent approaches use pre-trained embeddings to capture both semantic and syntactic information from legal texts. Models such as DeBERTa and BigBird have reported strong results on the ILDC dataset, achieving accuracies of up to 80%. In related work, [69] introduce a semantic role labeling model for Indian legal judgments that combines Named Entity Replacement with Graph Attention Networks to generate context-aware sentence representations. The proposed approach outperforms earlier methods and supports downstream tasks such as judgment analysis and summarization, enabling more efficient legal decision-making.

Recent work has explored a range of advanced strategies for improving the classification of Indian legal judgments. For instance, [70] integrates transformer-based

representations with summarization methods and a hierarchical BiLSTM CRF architecture to identify salient sentences, reporting gains in classification performance. In a related study, [71] conducted a comparative evaluation of several transformer models–including BERT, XLNet, RoBERTa, DeBERTa, ELECTRA, and BigBird on the ILDC dataset, demonstrating the effectiveness of contemporary NLP techniques for legal judgment prediction in the Indian setting.

Approaching the problem from a different perspective, [72] employs topic modeling based on Latent Dirichlet Allocation (LDA) to classify Indian legal judgments, with improvements observed in both classification accuracy and topic-level retrieval. To mitigate challenges related to class imbalance and feature complexity, [73] proposed a hybrid framework that combines feature engineering with machine and deep learning models, achieving accuracies of up to 98% when modern language model embeddings are incorporated.

In addition to classification, recent research has addressed the problem of legal text summarization. For example, [65] investigates abstractive summarization techniques and large language models for summarizing complex legal judgments drawn from Indian, UK, and U.S. jurisdictions. Their findings suggest that, although such models generally outperform extractive approaches, they often require human supervision to address issues related to hallucination and inconsistency. In a closely related line of work, [74] examines the identification of rhetorical roles within legal case documents, a task that supports downstream applications such as summarization and judgment prediction. The proposed MARRO framework, which combines multi-task learning with multi-headed attention mechanisms, reports strong results on datasets from the Indian Supreme Court.

2.4 Summary and research gaps

Legal text analysis has steadily progressed, with early studies [17, 30] laying the groundwork for processing legal documents. Subsequent research has advanced areas such as named entity recognition, knowledge graph construction, and legal sentence boundary detection [28, 75, 76], leading to more accurate interpretation and structuring of legal texts. Further progress in legal text analysis is reflected in studies on hybrid deep learning and rule-based methods [22, 25], which have expanded the analytical scope of legal information systems. At the same time, the use of pretrained language models and advanced NLP techniques [8, 77] has accelerated automation and improved legal text understanding. Integrating semantic analysis with knowledge graph construction has also enhanced the representation and interpretation of complex legal information [30]. Together, these efforts highlight the continued evolution of legal text analysis methodologies.

The pioneering study in structural classification of Indian legal judgments is performed by the researchers in [18]. The study uses zero-shot prompting with a combination of Mistral models to achieve a best accuracy of 61%. The study throws light on the efficacy of LLMs for legal document processing tasks. The work in this paper takes this aspect forward and performs a comprehensive analysis using transformer models for structural classification of legal judgments in the Indian legal scenario.

Despite advances in legal NLP, several task-specific research gaps remain in the structural classification of legal judgments. There is a notable shortage of annotated corpora

that support fine-grained tasks such as multiclass structural labeling of long legal judgments. Second, most existing studies focus on limited structural categories, whereas real-world judgments require multiclass structural classification across a larger and more complex set of sections. Moreover, systematic comparisons between traditional feature-based approaches, neural models, and contextual embedding techniques for this task are scarce, particularly for Indian legal texts. Additionally, current NLP methods often struggle to model long-range contextual dependencies and citation-driven discourse patterns that are intrinsic to judicial documents. The adoption of domain-specific representation learning techniques, particularly those based on legally adapted contextual embeddings such as Legal-BERT and Indian Legal-BERT, remains insufficiently explored in prior work on the classification of legal judgments.

3 Dataset distribution

The cornerstone of our analysis is a robust dataset provided by Indian Kanoon under a Non-Disclosure Agreement (NDA). The dataset underwent extensive data cleaning to ensure the integrity and applicability of the labels. The data curation process was meticulous, involving the alignment of various labels with the corresponding text segments from the judgments to ensure the accuracy of the input data for the subsequent modeling phases. The dataset's representativeness was ensured by including a diverse range of legal cases spanning various areas of law. The initial dataset had 28 classes, but the classes with equal to or fewer than 10 samples were pruned to maintain a standard of reliability and to facilitate a more focused and relevant analysis. Table 1 depicts the 13 low-frequency structural labels before class consolidation, which had less than or equal to 10 samples. So, the 13 classes with less than or equal to 10 samples were pruned, and the remaining dataset has a comprehensive collection of 478 judicial judgments, which have been systematically categorized into 15 classes. The dataset encapsulates approximately 6500 instances of annotated data, with a precise count of 6429 entries, forming the empirical basis for our experiments. Table 2 illustrates the process followed for data cleaning for structural class normalization and data consolidation. Table 3 gives the description of each class in the legal documents along with the number of instances of each class present in the dataset.

In Table 4, we present a comprehensive summary of text statistics across various classes of legal documents, including the average number of words per document, the

Table 1 Low-Frequency Structural Labels Before Class Consolidation

Structural Class	Number of Samples
intervenor	10
argument by plaintiff	6
obiter dicta	5
facts	4
applicant	2
defendant	2
negative	2
positive	1
argument by petitioner or appellant	1
argument by respondent or defendant	1
relied by party	1
parts of the structure	1
argument by intervenor	1

Table 2 Structural Class Normalization and Consolidation Process

Stage	Description
Initial Annotation	113 fine-grained structural labels extracted from Indian legal judgments, including petitioner/appellant arguments, respondent/defendant arguments, court discourse, interventions, and relied precedents.
Normalization	Labels were normalized to 54 classes by correcting spelling errors and unifying naming variations (e.g., *argument by petitioner*, *arguments of petitioner*, *argument by appellants*).
Semantic Merging	Semantically similar labels were merged, resulting in 28 consolidated structural classes representing distinct judgment components.
Final Selection	15 well-populated structural classes were retained for modeling, while 13 classes with fewer than 10 samples each were removed to ensure statistical reliability.

Table 3 Classwise Description and Number of Samples

Class	Description	Number of Samples
Precedent Analysis	Evaluating past legal decisions to determine their relevance and applicability to a current case	1585
Court Discourse	Interaction between judges and lawyers during a legal proceeding	901
Section Analysis	Segregation of the case to which probable sections each instance might belong to	878
Fact	Evidence that can be perceived by our senses	640
Issue	Question that must be resolved by the court in order to reach a decision	609
Conclusion	Final part of the document where the arguments are summarized	569
Argument by Respondent	Written or oral response to the petitioner's argument	399
Argument by Appellant	Written argument presented by the party who has initiated the appeal	359
Argument by Petitioner	Petitioner's version of the claims and issues in the case for their entitlement to the remedy they seek from the court	195
Respondent	Party against whom a legal action is being taken	90
Ratio	Rule of law that the court has applied to the facts of the case in order to reach its decision	62
Appellant	Party who is appealing a decision made by a lower court	46
Petitioner	Party who initiates a legal action by filing a petition or complaint	41
Judge	Responsible for ensuring that the proceedings are fair and impartial	31
Argument by Defendant	Presents the defendant's perspective on the legal issues and facts of the case	24

minimum and maximum word counts observed, and the size of the vocabulary used within each class. Classes range from 'Issue' and 'Court Discourse' to more specific categories like 'Argument by Petitioner' and 'Section Analysis', each exhibiting distinct linguistic characteristics. For instance, 'Court Discourse' documents tend to be lengthier with a higher average word count and a significantly larger vocabulary size, indicative of their complexity and the breadth of topics covered. In contrast, more focused document classes like 'Appellant' and 'Judge' show much lower word counts and vocabulary sizes, reflecting their targeted nature. This table serves as a quantitative overview of the linguistic diversity and complexity inherent in legal texts, highlighting the challenges and considerations in legal text analysis.

Along with the dataset acquired from Indian Kanoon, the research also involves a document including 5000 most frequently used phrases in the legal domain. This data is acquired after going through legal documents and formulating unigram, bigram, trigram, 4-gram and 5-gram groups of words. In our work, this phrase list is used only as

Table 4 Descriptive Statistics of Word Counts and Vocabulary Size for Each Legal Text Class

Class Name	Avg Words	Min Words	Max Words	Vocabulary Size
Precedent Analysis	176.59	4	5591	26212
Court Discourse	319.85	6	5237	24607
Section Analysis	229.78	3	3249	17380
Fact	287.24	8	1878	20861
Issue	90.63	3	1014	8814
Conclusion	118.05	2	909	8828
Argument by Respondent	173.59	6	1186	9860
Argument by Appellant	181.39	8	1306	9333
Argument by Petitioner	152.53	13	675	5544
Respondent	12.96	2	56	475
Ratio	206.26	22	949	3117
Appellant	11.11	3	30	257
Petitioner	17.78	3	119	337
Judge	6.19	3	18	107
Argument by Defendant	187.29	18	682	1358

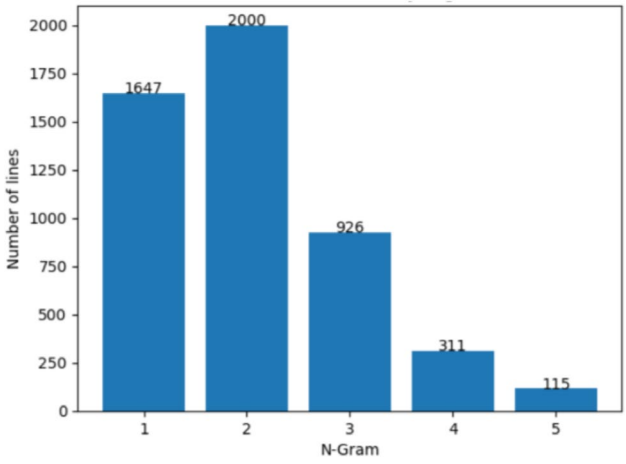


Fig. 1 n-gram Distribution of Phrases in Frequently used Words in Legal Domain

part of the feature construction process for traditional ML models. Specifically, these phrases are included in the n-gram vocabulary used to generate TF-IDF representations of the legal judgments, allowing the models to capture domain-specific legal expressions. The phrase list is not annotated with class labels and is not used for any supervised labeling task. It is not used for fine-tuning or training any large language or transformer-based models. The motivation for using this list comes from the data provider's interest in understanding how simple machine learning models leverage frequently occurring legal phrases and what impact these phrases have on classification performance. Figure 1 shows the distribution of these phrases in the dataset.

The dataset was shared as structured doctext files prepared by the legal team at Indian Kanoon, specifically for automated structural classification of Indian legal judgments. Because of this format, the original document layout is not preserved in a way that easily allows us to present visually annotated examples within the manuscript. In addition, a separate list of commonly used legal-domain phrases was provided and used only for constructing TF-IDF based features for the traditional machine learning models. These phrases were not annotated and were not meant for illustrative, document-level

examples. Our proposed approach operates as a wrapper on top of the baseline classification pipeline used by Indian Kanoon. Using the backend models developed in this work, the system automatically assigns structural labels to legal judgments across the 15 classes discussed in the manuscript. Table 5 depicts a few samples of data along with the corresponding judgment IDs.

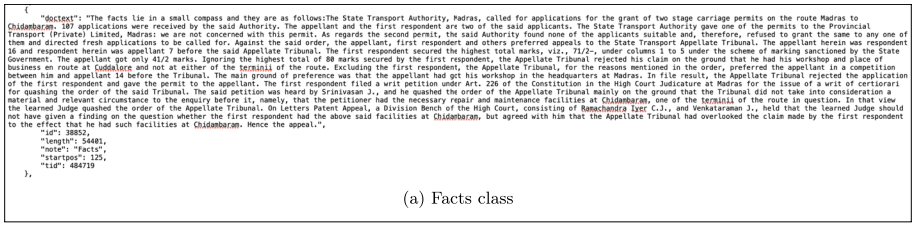
A few screenshots from the doctext file as provided by Indian Kanoon, depicting judgment ID and class information, are provided in Fig. 2 for the reader to get an understanding of the format of data provided for the structural classification of Indian legal judgments carried out in this work.

4 Methodology

This section describes various methods of converting text data into a format suitable for structural classification of legal judgments using various techniques such as n-gram based TF-IDF and contextual word embeddings and explains the building of different models. Figure 3 illustrates the various steps involved in the structural classification of legal documents. The process follows a multifaceted approach and begins with data preprocessing, including tasks like lowercasing, stop word removal, and addressing inconsistencies in class annotations. The data is then vectorized using techniques such as n-gram based TF-IDF and contextual word embeddings. Further to dimensionality reduction and augmentation, the next stage involves model building. Various learning architectures involving machine learning models (logistic regression, support

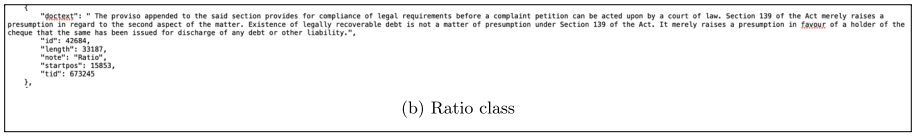
Table 5 Sample doctexts annotated by legal team from the IndianKanoon dataset

doctext	id	length	startpos	tid	note
The petitioner's case is that the shareholders of the Sholapur company have been subjected to discrimination visa vis the shareholders of other companies	38195	175939	5917	4354	Argument by petitioner
This is an application under article 32 of the Constitution seeking relief against alleged infringement of certain fundamental rights of the petitioner	38255	175939	21	4354	Issue
It has been held in a number of cases in the United States of America that no one except those whose rights are directly affected by a law can raise the question of the constitutionality of that law This principle has been very clearly stated by Hughes J. in McCabe v. Atchison(1)	57705	175939	899	4354	Precedent Analysis
I agree with the line of reasoning and the conclusion of Mr. Justice Mukherjea as regards the second point relating to the invalidity of the Act on the ground that it infringes article 14 of the Constitution and have nothing more to add. In my opinion therefore this petition fails and is dismissed with costs.	57703	175939	1529	4354	Conclusion
I have read the judgment prepared by Mr. Justice Mukherjea In respect of the arguments advanced to challenge the validity of the impugned Act under articles 31 and 19 of the Constitution of India, I agree with his line of reasoning and conclusion and have nothing more to add	57701	175939	410	4354	Courts Discourse



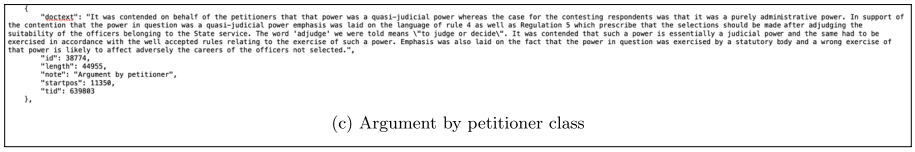
```
{
    "doctext": "The facts lie in a small compass and they are as follows:The State Transport Authority, Madras, called for applications for the grant of two stage carriage permits on the route Madras to
Chidambaram. 107 applications were received by the said Authority. The appellant and the first respondent are two of the said applicants. The State Transport Authority gave one of the permits to the Provincial
Transport (Private) Limited, Madras: we are not concerned with this permit. As regards the second permit, the said Authority found none of the applicants suitable and, therefore, refused to grant the same to any one of
them and directed fresh applications to be called for. Against the said order, the appellant, first respondent and others preferred appeals to the State Transport Appellate Tribunal. The appellant herein was respondent
16 and respondent herein was appellant 7 before the said Appellate Tribunal. The first respondent secured the highest total marks, viz., 71/2-, under columns 1 to 5 under the scheme of marking sanctioned by the State
Government. The appellant got only 41/2 marks. Ignoring the highest total of 80 marks secured by the first respondent, the Appellate Tribunal rejected his claim on the ground that he had his workshop and place of
business en route at Cuddalore and not at either of the terminii of the route. Excluding the first respondent, the Appellate Tribunal, for the reasons mentioned in the order, preferred the appellant in a competition
between him and appellant 14 before the Tribunal. The main ground of preference was that the appellant had got his workshop in the headquarters at Madras. In file result, the Appellate Tribunal rejected the application
of the first respondent and gave the permit to the appellant. The first respondent filed a writ petition under Art. 226 of the Constitution in the High Court Judicature at Madras for the issue of a writ of certiorari
for quashing the order of the said Tribunal. The said petition was heard by Srinivasan J., and he quashed the order of the Appellate Tribunal mainly on the ground that the Tribunal did not take into consideration a
material and relevant circumstance to the enquiry before it, namely, that the petitioner had the necessary repair and maintenance facilities at Chidambaram, one of the terminii of the route in question. In that view
the learned Judge quashed the order of the Appellate Tribunal. On Letters Patent Appeal, a Division Bench of the High Court, consisting of Ramachandra Iyer C.J., and Venkataraman J., held that the learned Judge should
not have given a finding on the question whether the first respondent had the above said facilities at Chidambaram, but agreed with him that the Appellate Tribunal had overlooked the claim made by the first respondent
to the effect that he had such facilities at Chidambaram. Hence the appeal.",
        "id": 38852,
        "length": 54481,
        "note": "Facts",
        "startpos": 125,
        "tid": 484719
    },
```


(a) Facts class



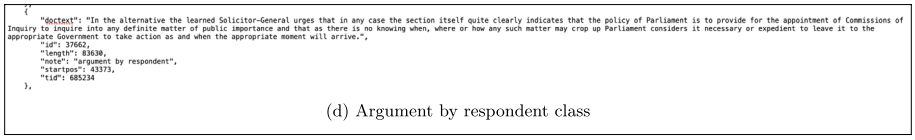
```
{
    "doctext": " The proviso appended to the said section provides for compliance of legal requirements before a complaint petition can be acted upon by a court of law. Section 139 of the Act merely raises a
presumption in regard to the second aspect of the matter. Existence of legally recoverable debt is not a matter of presumption under Section 139 of the Act. It merely raises a presumption in favour of a holder of the
cheque that the same has been issued for discharge of any debt or other liability.",
        "id": 42684,
        "length": 33187,
        "note": "Ratio",
        "startpos": 15853,
        "tid": 673245
    },
```


(b) Ratio class



```
{
    "doctext": "It was contended on behalf of the petitioners that that power was a quasi-judicial power whereas the case for the contesting respondents was that it was a purely administrative power. In support of
the contention that the power in question was a quasi-judicial power emphasis was laid on the language of rule 4 as well as Regulation 5 which prescribe that the selections should be made after adjudging the
suitability of the officers belonging to the State service. The word 'adjudge' we were told means \"to judge or decide\". It was contended that such a power is essentially a judicial power and the same had to be
exercised in accordance with the well accepted rules relating to the exercise of such a power. Emphasis was also laid on the fact that the power in question was exercised by a statutory body and a wrong exercise of
that power is likely to affect adversely the careers of the officers not selected.",
        "id": 38774,
        "length": 44955,
        "note": "Argument by petitioner",
        "startpos": 11350,
        "tid": 639803
    },
```


(c) Argument by petitioner class



```
{
    "doctext": "In the alternative the learned Solicitor-General urges that in any case the section itself quite clearly indicates that the policy of Parliament is to provide for the appointment of Commissions of
Inquiry to inquire into any definite matter of public importance and that as there is no knowing when, where or how any such matter may crop up Parliament considers it necessary or expedient to leave it to the
appropriate Government to take action as and when the appropriate moment will arrive.",
        "id": 37662,
        "length": 83630,
        "note": "argument by respondent",
        "startpos": 43373,
        "tid": 685234
    },
```


(d) Argument by respondent class

Fig. 2 Sample doctext files from the Indian Kanoon dataset depicting the judgment ID, start position and class details

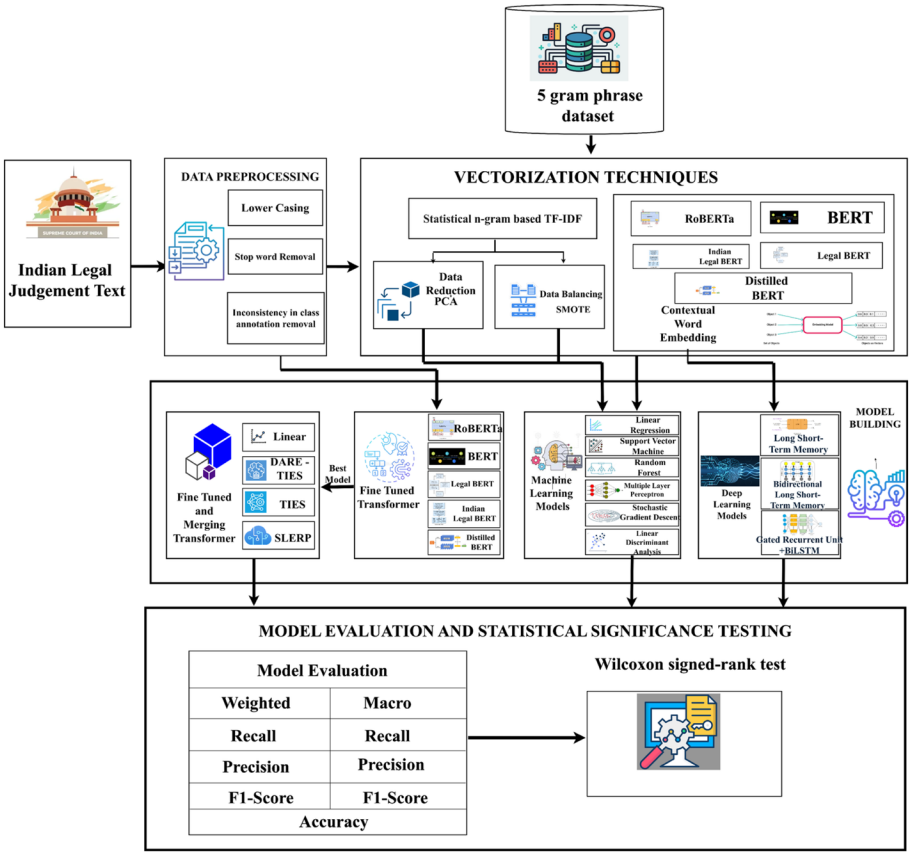


Fig. 3 Proposed Multifaceted Architecture for the Structural Classification of Indian Legal Judgments

vector machine, random forest, linear discriminant analysis, gradient boosting, stochastic gradient descent classifier, multilayer perceptron), deep learning models (LSTM, Bidirectional LSTM, GRU + Bidirectional LSTM), and fine-tuned transformer models are trained and the best models are merged. Finally, model evaluation is conducted using metrics such as weighted and macro recall, precision, F1-score, and accuracy. The detailed description of the various steps involved in the multifaceted process is discussed in the following subsections.

4.1 Data preprocessing

Prior to the modeling phase, the dataset underwent a thorough preprocessing process to ensure it was in a suitable format for analysis. This included basic cleaning steps such as lowercasing to unify class names like "Issue" and "issue," removing stopwords, and standardizing spacing. Additionally, inconsistencies in class annotations were addressed, combining terms such as "Argument by Respondent" and "Argumentrespondent" into a single class. These preprocessing techniques were crucial for enhancing the predictive capabilities of the model in the structural classification of Indian legal judgments. Followed by data preprocessing, various vectorization techniques are used to represent the Indian legal judgments in numerical vector format.

4.2 Vectorization techniques

Various range of vectorization techniques, like n-gram based TF-IDF and contextual word embeddings, are used to represent the structural Indian legal judgments. These techniques enabled the transformation of textual data into real value vector representation, capturing the actual meaning of the data. Subsequently, trained various machine learning models, deep learning models, and fine-tuned transformer models using these resultant vectors.

4.2.1 n-gram based TF-IDF

In the study, text data was transformed into numerical representations using n-gram based TF-IDF. Initially, a collection of 5000 phrases stored in a text file was utilized. The dataset consisted of 6429 legal documents. To convert these documents into numerical vectors, n-gram based TF-IDF was employed. N-gram based TF-IDF involves breaking down the text into different combinations of contiguous words, or "n-grams," such as single words, two-word (bigrams), three-word (trigrams) till six-word. This approach aimed to capture both individual keywords and meaningful phrases that hold significant context within legal documents. This method was particularly valuable in the legal domain, where understanding the context of specific phrases is critical for accurate analysis. Legal terminology often comprises phrases or combinations of words that convey specific meanings. By applying n-gram TF-IDF, these subtleties could be effectively captured and represented numerically. Term Frequency (TF) for n-gram t in a sample d is calculated as the number of times the n-gram appears in the sample divided by the total number of n-grams in that sample as shown in Eqn 1):

$$TF(t,d) = \frac{\text{Number of times n-gram t appears in document d}}{\text{Total number of n-grams in document d}} \tag{1}$$

Inverse Document Frequency (IDF) measures the importance of a phrase across the entire corpus of legal documents. IDF for a phrase t is calculated as the logarithm of the ratio of the total number of samples in the legal dataset (6429) to the number of samples containing the n-gram, with 1 added to the denominator to avoid division by zero, given in Eqn 2.

$$\mathrm{IDF}(t, D) = \log\left(\frac{\text{Total number of samples in legal structural dataset } D}{\text{Number of samples containing n-grams } t + 1}\right) \quad (2)$$

The TF-IDF score of a n-gram t in a sample d is then calculated by multiplying its TF and IDF scores,as shown in Eqn. 3:

$$TF - IDF(t, d, D) = TF(t, d) \times IDF(t, D) \quad (3)$$

For each legal document, the term frequency (TF) of individual phrases is computed, along with the inverse document frequency (IDF) across the full corpus. The resulting TF–IDF scores capture both the importance of a phrase within a specific document and its relevance across the entire collection of legal texts. This representation helps reflect the distinctive linguistic characteristics of legal documents. Applying n-gram based TF-IDF to the dataset produces a matrix with 6,429 rows, corresponding to individual samples, and 5,000 columns, representing the selected n-gram phrases. Each entry in the matrix contains the TF-IDF value of a given n-gram in a particular document, yielding a detailed numerical representation of the legal dataset for subsequent analysis.

4.2.2 Contextual word embeddings

In addition to the n-gram based TF-IDF technique, we also incorporated a diverse array of pre-trained word embedding models to represent the structural Indian legal judgments. These models, known for their ability to capture syntactic and semantic word relationships in dense vector spaces, were carefully selected to cover a broad spectrum of linguistic nuances pertinent to legal judgments. The models included:

- **BERT:** A pre-trained language model based on the transformer architecture that produces 768-dimensional contextual word embeddings. By modeling bidirectional context through masked language modeling, BERT effectively captures complex linguistic patterns. Its ability to be fine-tuned for downstream tasks has significantly advanced natural language understanding across domains, including legal text analysis [78].
- **DistilBert:** Lightweight version of BERT developed to reduce computational cost while preserving much of BERT's performance. It produces 768-dimensional embeddings and uses a knowledge distillation approach to learn from the larger BERT model. This balance between efficiency and accuracy makes DistilBERT suitable for tasks where computational resources are limited [79].
- **Roberta:** RoBERTa extends the BERT architecture by improving the pre-training process, including changes to training objectives, dataset size, and hyperparameter settings. It generates 768-dimensional contextual embeddings that capture complex language patterns more effectively, making it well suited for tasks involving domain-specific text, such as legal documents [80].

- **Legal BERT:** LegalBERT is a domain-adapted variant of BERT that has been further trained on large-scale legal corpora. It produces 768-dimensional embeddings that capture legal terminology and contextual dependencies more accurately than general-purpose language models. Through transfer learning and transformer-based representations, LegalBERT is well suited for legal NLP tasks requiring precise semantic understanding [81].
- **Indian Legal BERT:** Indian Legal BERT is specifically fine-tuned on Indian legal texts to model the linguistic and contextual characteristics of Indian jurisprudence. Like BERT, it generates 768-dimensional contextual embeddings using transformer architecture, but its training data and objectives are tailored to Indian legal documents. This specialization makes it particularly effective for tasks involving Indian court judgments and legal analysis [82].

Model embeddings were generated for Indian legal judgments using both n-gram based TF-IDF representations and contextual word embeddings. To address the issue of class imbalance, data augmentation was incorporated selectively. Specifically, SMOTE was applied only to the vectors obtained from the n-gram based TF-IDF representations, while the word embedding vectors were left unaltered. This choice is motivated by the fundamental differences between these two vectorization strategies.

In TF-IDF representations, each feature value reflects the relative importance of a specific term within a document. This structure allows SMOTE to generate meaningful synthetic samples by interpolating feature values while preserving term significance across classes. In contrast, word embedding vectors encode semantic and contextual relationships among words rather than explicit term importance. Applying data augmentation directly to such embeddings can distort these semantic relationships and adversely affect the quality of the learned representations. Consequently, data augmentation was restricted to the TF-IDF feature space to ensure balanced training data while maintaining the integrity of the semantic information captured by word embeddings.

4.3 Data augmentation

In the original Indian legal judgments, we observed significant class imbalance as shown in Fig. 4, with certain classes having a disproportionately low number of samples compared to others. For instance, classes such as 'Appellant', 'Argument by Defendant', 'Petitioner', and 'Judge' had considerably fewer instances compared to dominant classes like 'Court Discourse' and 'Precedent Analysis'. This imbalance could potentially lead to biased model predictions, where the model tends to favor the majority classes due to their higher representation in the dataset.

To address this issue and ensure fair representation of all classes in our model training process, we employed the Synthetic Minority Over-sampling Technique (SMOTE). After applying SMOTE to our dataset, we observed a significant increase in the number of samples for the minority classes, resulting in a more balanced distribution across all classes. The resampled dataset after applying SMOTE has a shape of (23775, 5000), with each class now containing an equal number of samples (1585 samples per class). This balanced class distribution enhances the reliability and fairness of our model.

After n-gram based TF-IDF vectorization and the application of SMOTE, the feature set of our Indian legal judgments became quite large, comprising around 5000 features. Such a high-dimensional feature space can lead to the curse of dimensionality, making

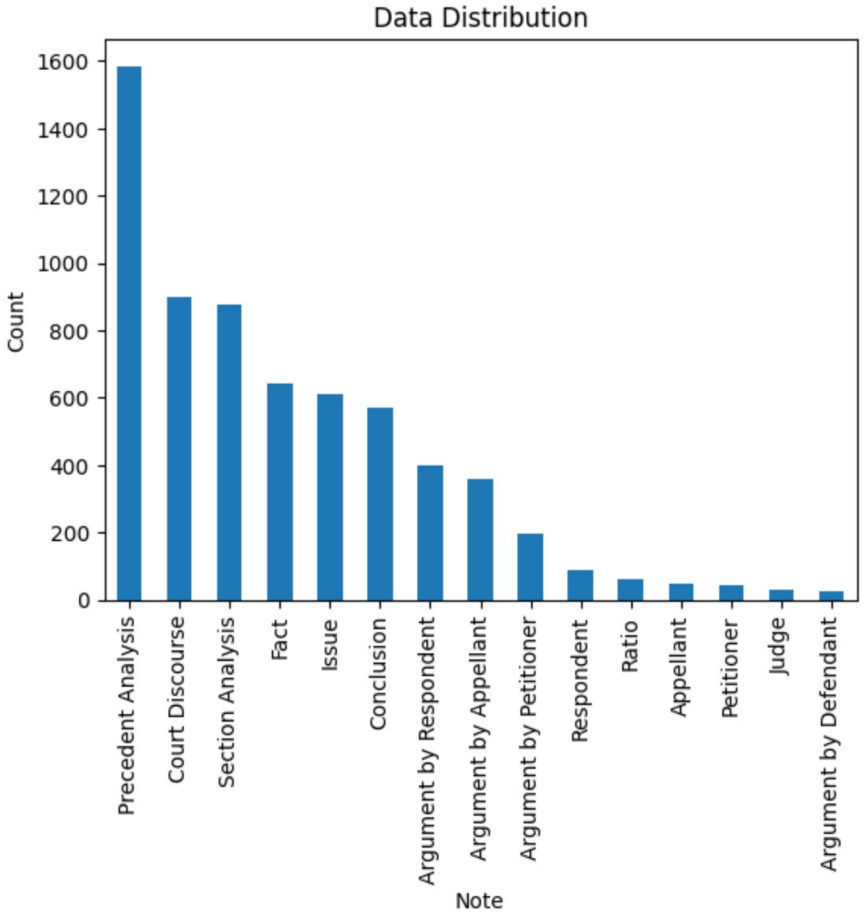


Fig. 4 Distribution of instances across 15 classes in the dataset

the data challenging to manage and interpret. To address this issue, we implemented feature reduction techniques, specifically Principal Component Analysis (PCA), which is discussed in the subsequent sections. On the other hand, word embeddings inherently possess a lower dimensionality, typically around 768 features, which is considerably more manageable. Therefore, we did not apply any feature reduction techniques to the word embeddings.

4.4 Feature reduction & model building

We applied PCA on n-gram based TF-IDF vectors for feature reduction to 100, 200, 300, 400, and 500 dimensions. This allowed us to explore how different levels of feature reduction affect our model performance. After reducing the features, we trained machine learning models with each of the above feature combinations respectively to find the optimal number of features for accurate predictions and thereby reducing the complexity of the dataset. Following the vectorization techniques and PCA on n-gram based TF-IDF vectors, we proceeded to train our dataset using a variety of learning models.

4.4.1 Machine learning

Implemented various ML models as mentioned here to classify structural Indian legal judgments.

- **Logistic Regression (LR):** A simple yet effective algorithm for binary classification, estimating the probability of a binary outcome based on predictor variables by fitting a logistic (sigmoid) function.
- **Support Vector Machines (SVM):** Capable of performing classification and regression tasks by finding the optimal hyperplane that separates data points of different classes in the feature space, maximizing the margin between classes. The Radial Basis Function (RBF) kernel was employed for the SVM model to capture the non-linear relationships within the dataset.
- **Random Forest (RF):** Constructs multiple decision trees and outputs the mode of classes or mean prediction. In this study, RF model was used with default parameters.
- **Linear Discriminant Analysis (LDA):** Used for dimensionality reduction and classification by projecting features onto a lower-dimensional subspace while maximizing class separation.
- **Gradient Boosting (GB):** Ensemble learning technique that sequentially combines weak learners by fitting each new model to the residual errors of the previous model. Renowned for its high predictive accuracy and ability to capture complex relationships.
- **Stochastic Gradient Descent Classifier(SGDC):** Updates model parameters using a single random sample or a small subset of training data at a time, making it suitable for large-scale datasets and online learning scenarios due to its computational efficiency.
- **Multilayer Perceptron (MLP):** Feedforward artificial neural network capable of modeling complex relationships between inputs and outputs, making them adept at capturing nuanced patterns in the data.

The models are trained on three distinct types of feature representations: n-gram based TF-IDF after PCA, n-gram based TF-IDF with SMOTE after PCA and word embeddings. By training these ML models on these three types of representations, we aimed to compare their performance and determine the most suitable model for classification of Indian legal judgments.

4.4.2 Deep learning models

Deep learning models are applied directly to word embeddings because word embeddings capture semantic relationships and provide dense representations. TF-IDF matrices tend to have high dimensionality and sparsity, posing computational challenges. DL models were specifically employed for their superior performance in capturing the sequential and contextual nature of text data. The selected architectures included:

- **Long Short-Term Memory (LSTM):** LSTM networks are effective in modeling sequential data and long-term dependencies in legal texts. We use a single LSTM layer with 256 units, followed by three dense layers of 512, 256, and 128 units with ReLU activation, and a final softmax layer with 15 units for classification.
- **Bidirectional LSTM:** This model enhances standard LSTMs by processing sequences in both forward and backward directions, allowing it to capture richer contextual information from legal texts. The architecture uses a bidirectional Keras wrapper with LSTM layers of 512 and 256 units. The resulting output is passed through a flattening layer, followed by three dense layers with 512, 256, and 128

units using ReLU activation. Finally, a softmax output layer with 15 units is used for multiclass classification.

- **Gated Recurrent Unit (GRU) + Bidirectional LSTM:** The combination of GRU and bidirectional LSTM architectures balances efficiency with strong contextual understanding for legal text analysis. All DL models use advanced embeddings, while TF-IDF features are excluded due to their static nature. The best model consists of three bidirectional LSTM layers (768, 512, and 256 units) with dropout for regularization, followed by dense layers of 512, 256, and 128 units with ReLU activation. A softmax output layer with 15 units performs multiclass classification, and the models are trained for up to 100 epochs.

4.4.3 Fine-tuning and merging of BERT-based transformer models

This study fine-tunes several BERT-based transformer models on Indian legal judgments to improve structural classification. The models include BERT, DistilBERT, RoBERTa, LegalBERT, and Indian LegalBERT, selected for their complementary strengths. BERT provides a strong baseline, DistilBERT offers faster and more efficient inference, and RoBERTa is better suited for handling noisy legal text. LegalBERT captures domain-specific legal language, while Indian LegalBERT plays a key role due to its specialization in Indian legal texts. To further improve performance, we merge fine-tuned LegalBERT and Indian LegalBERT models using advanced interpolation techniques. This approach combines their complementary representations, allowing the final model to leverage the strengths of both and produce more robust and accurate predictions.

The merging strategies used for this purpose are described below:

- **Spherical Linear Interpolation (SLERP)**: Input vectors from both models were first normalized to unit length, and the angular distance between corresponding vectors was computed using their dot product. Layer-wise interpolation was then performed with predefined weights: [0, 0.5, 0.3, 0.7, 1] for self-attention layers, while for the MLP layers, we used [1, 0.5, 0.7, 0.3, 0]. These weights guided the interpolation process, producing a merged model that balances the complementary representations of LegalBERT and Indian LegalBERT.
- **Trim, Elect Sign & Merge (TIES)**: Parameters with minimal variation were pruned to reduce noise, and sign conflicts were resolved by retaining stable, consistent parameters. Both models were equally weighted (0.5), with a density parameter set to 0.53 during merging.
- **Drop and Rescale TIES (DARE-TIES)**: The fine-tuned LegalBERT and Indian LegalBERT models were merged using a combination of Drop and Rescale (DARE) and TIES, where low-impact parameters were removed and the remaining parameters were rescaled to maximize their contribution to performance.
- **Linear**: This approach merged the fine-tuned LegalBERT and Indian LegalBERT models by computing a weighted average of their parameters, with equal weights (1.0) assigned to both models. To prevent parameter shrinkage, particularly in deeper layers, appropriate scaling was applied during the merging process.

4.5 Model evaluation

After training various models on the dataset, a thorough evaluation was conducted using multiple metrics:

- **Accuracy(Acc):** This metric assesses the overall correctness of the model's predictions, indicating the proportion of correctly classified instances out of the total instances.
- **Precision(P):** Precision focuses on the accuracy of positive predictions, measuring the proportion of correctly predicted positive instances out of all instances predicted as positive.
- **Recall(R):** Also known as sensitivity, recall gauges the model's ability to capture all positive instances by calculating the proportion of correctly predicted positive instances out of all actual positive instances.
- **F1 Score(F1):** The F1 score, the harmonic mean of precision and recall, offers a balanced measure of a model's performance.

Two common evaluation methods employed are:

- **Weighted Results:** Weighted averaging considers class imbalance in the dataset by assigning weights based on class frequency, ensuring proportional evaluation.
- **Macro Results:** Macro averaging computes metrics independently for each class and takes the average across all classes, treating all classes equally.

In this study, we primarily rely on macro-averaged precision, recall, and F1-score to evaluate model performance, as these metrics are particularly well suited for multi-class classification problems with imbalanced class distributions. Unlike accuracy or instance-weighted measures, macro-averaged metrics treat all classes equally, ensuring that performance on frequently occurring classes does not overshadow that of under-represented ones [83, 84]. Specifically, macro-averaging computes precision, recall, and F1-score independently for each class and then averages the results across all categories. This approach makes the evaluation insensitive to class frequency and provides a more reliable assessment of performance on minority classes. By emphasizing macro-averaged metrics, our evaluation framework offers a more balanced and discriminative view of per-class behavior, in line with best practices for imbalanced classification and prior studies that favor macro-level evaluation even in oversampled settings [84–86]. Notably, the macro F1-score is widely regarded as one of the most appropriate performance measures for imbalanced datasets [85].

4.6 Experimental setup

In model training, the focus was on improving performance by fine-tuning hyperparameters. These adjustments were crucial for effectively training the models and enhancing their ability to make accurate predictions on unseen data. Table 6 outlines the parameters and their corresponding values. The training–testing split ratio was set to 80:20 for DL models and 5-fold cross validation was conducted for ML models.

The hardware used for training ML and DL models was T4 GPU on Google Colab instance which has 16GB RAM with 2560 CUDA cores and standard CPU settings. For training Transformer based models that require higher computational resources, we have used P100 GPU paired with 16GB HBM2 RAM and 3584 CUDA cores along with standard CPU settings provided by Google Cloud which was connected to a Google Colab notebook instance.

Table 6 Hyperparameters for the models

Parameter	Value
Machine Learning	
Epochs	25-250 depending on requirement
Kernels used in SVM	rbf
Kernel used in XGB	tree_method='gpu_hist'
Random forest number of trees	100
MLPClassifier	hidden_layer_sizes=(100,), activation='relu', solver='adam'
SGDClassifier	loss='hinge', penalty='l2'
Cross validation	5-Fold cross validation
Sampling technique	stratified sampling
Logistic regression penalty	L2 penalty
Packages	Scikit Learn, TensorFlow, Keras, Pandas
PCA size	100,200,300,400,500
Embedding size	BERT based:768
Deep Learning	
Optimizer	Adam
Epochs	25-250 depending on requirement
Learning Rate	initial 0.001, min 1e-10
Loss	Categorical Cross Entropy, hinge
Batch Size	16 or 32 based on GPU availability
Embedding size	BERT based:768
Metrics	F1 Score, Accuracy and Loss
Callbacks	ReduceLRonPlateau and Model Checkpoint
Training testing split	80:20
Packages	TensorFlow, Keras, PyTorch, Pandas
Fine Tuned Transformer Models	
Learning Rate	5e-05
Train Batch Size	8
Eval Batch Size	8
Optimizer	Adam with betas=(0.9,0.999) and epsilon=1e-08
LR Scheduler Type	linear
Num Epochs	15
Merged Transformer Models	
SLERP	self_attn=[0, 0.5, 0.3, 0.7, 1], mlp=[1, 0.5, 0.7, 0.3, 0]
TIES	normalize=false, int8_mask=true, density=0.5, weight=0.5
DARE-TIES	int8_mask=true, density=0.53, weight=1
Linear	weight=1

5 Experimental results & analysis

This section looks at the results of analyzing structural judgment classification using learning based models. Performance analysis of various models was performed and the classification report of the best performing model has been provided for that particular experiment technique. The following sections give detailed insights into the performance of each model and the inferences. The results are categorised based on the following analysis paradigms as given below:

- ML Models with n-gram based TF-IDF and PCA
- ML models with n-gram based TF-IDF and PCA with SMOTE
- ML Models with contextual Word Embeddings
- DL models with Contextual Word Embeddings
- Fine-tuned BERT based transformer models and merged best models

Each of the results and pertaining analysis are elaborated in the following subsections.

5.1 ML Models with n-gram based TF-IDF and PCA

The application of PCA on TF-IDF embeddings reduced dimensionality to 100, 200, 300, 400 and 500 features. The choice of PCA as the feature reduction method was driven by its consistent superiority to other methods. Increasing PCA components in model consistently exhibited notable improvements in both accuracy and F1-Scores. Detailed data about each model and its performance for n-gram based TF-IDF and PCA can be found in Table 7. It is worth noting that LDA was notably outstanding among all other performers attaining the highest F1 score and best accuracy of 0.72 with 500 features. These findings not only underscore the critical role played by feature reduction in mitigating dimensionality issues but also reveal effectiveness of LDA when dealing with reduced feature space leading to optimal model outcomes.

The classification report presented in Table 8 provides insights into the performance of LDA for n-gram based TF-IDF and PCA-500 across various classes. Classes such as "Conclusion" and "Precedent Analysis" exhibit relatively higher F1-scores (0.90 and

Table 7 Results of ML models with n-gram based TF-IDF and PCA

MODEL	PCA-Value	Weighted			Macro			Acc
		R	P	F1	R	P	F1	Acc
LR	pca-100	0.67	0.67	0.66	0.53	0.64	0.56	0.67
SVM		**0.72**	**0.71**	**0.71**	**0.61**	**0.67**	**0.63**	**0.72**
RF		0.65	0.65	0.64	0.55	0.62	0.57	0.65
LDA		0.67	0.70	0.68	0.58	0.62	0.58	0.67
GB		0.64	0.64	0.64	0.53	0.57	0.54	0.64
SGDC		0.68	0.67	0.66	0.58	0.69	0.6	0.68
MLP		0.69	0.68	0.68	0.63	0.65	0.64	0.69
LR	pca-200	0.70	0.69	0.69	0.55	0.67	0.59	0.70
SVM		**0.72**	**0.72**	**0.71**	**0.60**	**0.67**	**0.63**	**0.72**
RF		0.62	0.62	0.6	0.52	0.60	0.54	0.62
LDA		0.69	0.71	0.70	0.59	0.64	0.6	0.69
GB		0.64	0.64	0.64	0.52	0.56	0.54	0.64
SGDC		0.70	0.69	0.69	0.61	0.69	0.62	0.70
MLP		0.64	0.69	0.69	0.63	0.66	0.64	0.69
LR	pca-300	0.70	0.70	0.69	0.56	0.67	0.59	0.70
SVM		**0.72**	**0.72**	**0.71**	**0.61**	**0.67**	**0.63**	**0.72**
RF		0.63	0.63	0.59	0.51	0.62	0.54	0.61
LDA		0.69	0.71	0.7	0.6	0.65	0.61	0.69
GB		0.63	0.63	0.63	0.52	0.54	0.52	0.63
SGDC		0.7	0.69	0.69	0.61	0.69	0.62	0.7
MLP		0.67	0.68	0.67	0.63	0.66	0.64	0.67
LR	pca-400	0.71	0.7	0.7	0.57	0.68	0.61	0.71
SVM		**0.72**	**0.72**	**0.71**	**0.62**	**0.66**	**0.63**	**0.72**
RF		0.63	0.63	0.58	0.5	0.63	0.52	0.61
LDA		0.7	0.72	0.71	0.63	0.67	0.64	0.7
GB		0.63	0.63	0.63	0.53	0.56	0.52	0.63
SGDC		0.71	0.71	0.7	0.63	0.72	0.65	0.71
MLP		0.66	0.63	0.66	0.62	0.63	0.62	0.66
LR	pca-500	0.7	0.69	0.68	0.55	0.66	0.59	0.7
SVM		0.72	0.72	0.72	0.61	0.67	0.63	0.72
RF		0.61	0.61	0.57	0.49	0.62	0.51	0.6
LDA		**0.72**	**0.73**	**0.72**	**0.69**	**0.64**	**0.65**	**0.72**
GB		0.62	0.62	0.62	0.54	0.56	0.55	0.62
SGDC		0.68	0.67	0.66	0.58	0.69	0.6	0.68
MLP		0.66	0.65	0.65	0.6	0.63	0.61	0.65

Table 8 Classification Report of LDA for n-gram based TF-IDF and PCA-500

Class	Recall	Precision	F1-score	Support
Precedent Analysis	0.79	0.85	0.82	317
Court Discourse	0.68	0.54	0.60	180
Section Analysis	0.78	0.74	0.76	176
Fact	0.75	0.67	0.71	128
Issue	0.76	0.83	0.79	122
Conclusion	0.84	0.96	0.90	114
Argument by Respondent	0.51	0.61	0.56	80
Argument by Appellant	0.47	0.61	0.53	72
Argument by Petitioner	0.44	0.40	0.41	39
Respondent	0.67	1.00	0.80	18
Ratio	0.25	0.18	0.21	12
Appellant	0.67	1.00	0.80	9
Petitioner	0.88	0.78	0.82	8
Judge	0.50	1.00	0.67	6
Argument by Defendant	0.60	0.25	0.35	5
Accuracy	0.72			1286
Macro Avg	0.64	0.69	0.65	1286
Weighted Avg	0.72	0.73	0.72	1286

0.82, respectively), precision, and recall values, indicating effective classification performance. Conversely, classes like “Argument by Defendant” and “Ratio” demonstrate lower F1-scores (0.35 and 0.21, respectively) along with lower precision and recall values, suggesting challenges in accurately identifying these classes. Classes with larger support, such as “Precedent Analysis" and “Court Discourse" tend to have more balanced performance metrics, while classes with smaller support, such as “Ratio" and “Appellant" may suffer from imbalanced data or inherent complexity in classification. Overall, the model achieves an accuracy of 0.72 and shows balanced performance across macro and weighted averages, with F1-scores of 0.65 and 0.72, respectively. Further analysis and potentially more data for classes with lower support could improve the model's overall performance.

5.2 ML models with n-gram based TF-IDF and PCA with SMOTE

SMOTE when used in conjunction with PCA significantly enhanced model performance, addressing the class imbalance effectively. Model and its performance with each level of PCA-SMOTE can be found in Table 9. The best F1 score and accuracy of 0.942 is obtained with MLP on 300 and 500 dimensional features. This advanced data augmentation technique outperforms all the other ML and DL experiments performed on different embeddings and TF-IDF with feature reduction.

In Table 9, showcasing the performance of various ML models with PCA-SMOTE, RF consistently outperforms other models across different PCA dimensions with F1-scores, precision, recall, and accuracy all consistently at 0.93. SVM also demonstrates strong performance with F1-scores, precision, recall, and accuracy consistently at 0.9. On the other hand, LR and LDA show slightly lower but commendable performance, with F1-scores hovering around 0.87 and 0.86 respectively. SGDC, and MLP exhibit moderate performance, with F1-scores ranging from 0.84 to 0.86 across different PCA dimensions.

In Table 10, the classification report of MLP model for TF-IDF with PCA-200 demonstrates strong performance across various classes. Classes such as 'Respondent,' 'Judge,' 'Argument by Defendant,' 'Petitioner,' 'Appellant,' and 'Ratio' exhibit perfect F1-scores,

Table 9 Results of ML models with n-gram based TF-IDF and PCA with SMOTE

MODEL	PCA-Value	Weighted			Macro			
		R	P	F1	R	P	F1	Acc
LR	smote-pca-100	0.81	0.82	0.81	0.81	0.82	0.82	0.82
SVM		0.83	0.83	0.83	0.83	0.83	0.83	0.83
RF		**0.92**	**0.92**	**0.92**	**0.92**	**0.92**	**0.92**	**0.92**
LDA		0.78	0.78	0.78	0.78	0.78	0.78	0.78
GB		0.86	0.86	0.86	0.86	0.86	0.86	0.86
SGDC		0.78	0.78	0.79	0.78	0.78	0.79	0.79
MLP		0.91	0.91	0.91	0.91	0.91	0.91	0.91
LR	smote-pca-200	0.84	0.85	0.85	0.85	0.85	0.85	0.85
SVM		0.87	0.87	0.87	0.87	0.87	0.87	0.87
RF		0.93	0.93	0.93	0.93	0.93	0.93	0.93
LDA		0.82	0.83	0.81	0.82	0.83	0.81	0.816
GB		0.86	0.86	0.86	0.84	0.85	0.84	0.86
SGDC		0.83	0.83	0.83	0.83	0.83	0.83	0.83
MLP		**0.94**	**0.94**	**0.94**	**0.94**	**0.94**	**0.94**	**0.94**
LR	smote-pca-300	0.86	0.86	0.86	0.86	0.86	0.86	0.86
SVM		0.88	0.88	0.88	0.88	0.88	0.88	0.88
RF		0.93	0.93	0.93	0.93	0.93	0.93	0.93
LDA		0.84	0.85	0.84	0.84	0.85	0.84	0.84
GB		0.85	0.85	0.85	0.82	0.84	0.82	0.85
SGDC		0.85	0.85	0.85	0.85	0.85	0.85	0.85
MLP		**0.94**	**0.94**	**0.94**	**0.94**	**0.94**	**0.94**	**0.94**
LR	smote-pca-400	0.87	0.87	0.88	0.87	0.88	0.88	0.88
SVM		0.89	0.89	0.89	0.89	0.89	0.89	0.89
RF		0.93	0.93	0.93	0.93	0.93	0.93	0.93
LDA		0.86	0.87	0.85	0.86	0.87	0.85	0.85
GB		0.85	0.85	0.84	0.84	0.86	0.83	0.85
SGDC		0.86	0.87	0.87	0.86	0.87	0.87	0.87
MLP		**0.94**	**0.94**	**0.94**	**0.94**	**0.94**	**0.94**	**0.94**
LR	smote-pca-500	0.88	0.88	0.88	0.88	0.88	0.88	0.88
SVM		0.90	0.90	0.90	0.90	0.90	0.90	0.90
RF		0.93	0.93	0.93	0.93	0.93	0.93	0.93
LDA		0.87	0.87	0.86	0.87	0.87	0.86	0.86
GB		0.84	0.85	0.84	0.84	0.86	0.83	0.84
SGDC		0.87	0.87	0.87	0.87	0.87	0.87	0.87
MLP		**0.94**	**0.94**	**0.94**	**0.94**	**0.94**	**0.94**	**0.94**

precision, and recall, indicating flawless classification performance. Additionally, classes like 'Argument by Petitioner,' 'Argument by Appellant,' 'Conclusion,' and 'Issue' also show very high F1-scores, precision, and recall, suggesting accurate classification with minimal misclassifications. However, classes such as 'Precedent Analysis,' 'Court Discourse,' 'Fact,' and 'Section Analysis' exhibit slightly lower F1-scores, precision, and recall compared to other classes, although still performing reasonably well. The high performance across most classes indicates the robustness and effectiveness of the MLP model trained on TF-IDF data with PCA-200. This performance highlights the capability of the model to capture intricate patterns and nuances in the text data, enabling accurate classification even with dimensionality reduction to PCA-200.

5.3 ML Models with contextual Word Embeddings

The investigation into ML models showcased a discernible trend where SVM and LDA consistently outperformed other models across most embeddings. These findings

Table 10 Classification Report of MLP for n-gram based TF-IDF and PCA-200 with SMOTE

Class	Recall	Precision	F1-score	Support
Precedent Analysis	0.79	0.85	0.82	317
Court Discourse	0.78	0.81	0.79	317
Section Analysis	0.87	0.90	0.88	317
Fact	0.88	0.90	0.89	317
Issue	0.94	0.90	0.92	317
Conclusion	0.96	0.94	0.95	317
Argument by Respondent	0.96	0.96	0.96	317
Argument by Appellant	0.98	0.94	0.96	317
Argument by Petitioner	0.98	0.95	0.97	317
Respondent	1.00	1.00	1.00	317
Ratio	1.00	0.98	0.99	317
Appellant	0.94	1.00	0.97	317
Petitioner	1.00	0.94	0.97	317
Judge	1.00	1.00	1.00	317
Argument by Defendant	1.00	0.99	1.00	317
Accuracy	0.94			4755
Macro Avg	0.94	0.94	0.94	4755
Weighted Avg	0.94	0.94	0.94	4755

suggest a strong suitability of these models for complex semantic tasks in legal documents. LegalBERT with SVM offered the highest weighted F1 scores (0.69) and logistic regression gave the highest accuracy of 0.7. Detailed data on the performance of each ML model with embedding can be found in Table 11.

The best performing model employed LDA classifier, along with pre-trained Indian legal BERT embeddings, as evidenced by the contextual word embeddings with ML models. The corresponding classification report is provided in Table 12.

Indian legal BERT with LDA model demonstrates considerable effectiveness in legal judgments classification with an overall accuracy of 0.72, indicating a strong ability to correctly predict 72% of the sample texts. The model shows a high degree of precision in identifying 'Judge' and 'Respondent' classes, which suggests an excellent capability in recognizing these specific categories. However, it struggles with the 'Argument by Defendant' class, evidenced by a low F1-score of 0.12, pointing to difficulties in capturing the nuances of this category. The macro averages for precision, recall, and F1-score hover around 0.62–0.64, reflecting moderate overall performance across all classes. Among the classes, those with higher support values give higher F1-scores compared to that with lower support. However, certain exceptions are noticed for 'Respondent' and 'Judge' classes. These classes provide higher F1-scores (0.91 and 0.97) even with lesser support. This points to the fact that the testing sample embeddings are better reflecting the nuances of the class in this case. However, the 'Argument by Petitioner' and 'Argument by Defendant' provide lower F1-scores due to the presence of out of vocabulary words in the testing subset in the classification pipeline. The substantial support count of 1,286 instances provides a robust dataset for analysis, ensuring that the performance metrics are reliable.

5.4 DL models with contextual word embeddings

The study of deep learning models has uncovered promising outcomes with the integration of bi-LSTM and GRU structures, particularly when combined with Indian legal BERT. The combined model yielded an accuracy of 0.76 and F1 score of 0.75 as shown

Table 11 Results of Contextual Word Embeddings with ML Models

Embedding	Model	Weighted			Macro			
		R	P	F1	R	P	F1	Acc
BERT	LR	0.69	0.68	0.68	0.60	0.62	0.61	0.69
	SVM	0.67	0.67	0.67	0.60	0.61	0.62	0.67
	RF	0.60	0.57	0.56	0.46	0.51	0.47	0.60
	LDA	0.68	0.69	0.69	0.60	0.63	0.61	0.68
	GB	0.64	0.62	0.62	0.51	0.55	0.53	0.64
	SGDC	0.66	0.65	0.65	0.57	0.58	0.57	0.66
	MLP	0.66	0.65	0.65	0.59	0.60	0.60	0.66
Distilled Bert	LR	0.68	0.66	0.67	0.59	0.62	0.60	0.68
	SVM	0.68	0.67	0.68	0.59	0.61	0.58	0.68
	RF	0.60	0.56	0.56	0.44	0.51	0.45	0.60
	LDA	0.68	0.69	0.69	0.59	0.62	0.60	0.68
	GB	0.64	0.63	0.63	0.48	0.54	0.50	0.64
	SGDC	0.65	0.65	0.65	0.56	0.60	0.57	0.65
	MLP	0.69	0.69	0.69	0.65	0.64	0.64	0.69
Roberta	LR	0.65	0.63	0.63	0.49	0.53	0.50	0.65
	SVM	0.67	0.67	0.68	0.60	0.62	0.59	0.68
	RF	0.61	0.56	0.56	0.42	0.47	0.42	0.61
	LDA	0.69	0.69	0.69	0.58	0.6	0.59	0.69
	GB	0.62	0.64	0.63	0.46	0.44	0.43	0.62
	SGDC	0.57	0.58	0.58	0.45	0.52	0.46	0.57
	MLP	0.69	0.69	0.69	0.63	0.61	0.61	0.69
Legal Bert	LR	0.70	0.69	0.69	0.57	0.6	0.58	0.7
	SVM	0.70	0.69	0.69	0.59	0.64	0.6	0.7
	RF	0.62	0.59	0.59	0.41	0.51	0.41	0.62
	LDA	0.69	0.7	0.67	0.57	0.62	0.59	0.69
	GB	0.67	0.64	0.64	0.48	0.52	0.49	0.67
	SGDC	0.66	0.65	0.6	0.54	0.6	0.55	0.66
	MLP	0.69	0.68	0.68	0.58	0.6	0.59	0.69
Indian Legal BERT	LR	0.72	0.71	0.72	0.59	0.61	0.59	0.72
	SVM	0.73	0.72	0.72	0.58	0.63	0.59	0.73
	RF	0.64	0.61	0.60	0.43	0.49	0.42	0.64
	LDA	**0.72**	**0.73**	**0.72**	**0.61**	**0.64**	**0.62**	**0.72**
	GB	0.68	0.66	0.66	0.51	0.57	0.52	0.68
	SGDC	0.68	0.69	0.69	0.57	0.59	0.57	0.68
	MLP	0.72	0.71	0.71	0.59	0.61	0.59	0.72

in Table 13. BiLSTM considers both past and future context, while the GRU structure efficiently processes sequential information leading to better results. A detailed comparative analysis, depicted in Table 13, elucidates the strengths and performance metrics of these architectures, providing valuable insights. From Table 13, it can be observed that bi-LSTM+GRU outperforms other models, especially when used with embeddings like BERT and LegalBERT. However, with RoBERTa, performance is weaker in macro F1. Similar trend exists for precision and recall.

Figure 5 provides the loss curves for the model at each epoch for both training and validation for both Bi-LSTM and Bi-LSTM+GRU model with Indian Legal BERT embedding and Fig. 6 gives a glimpse into the model F1 score.

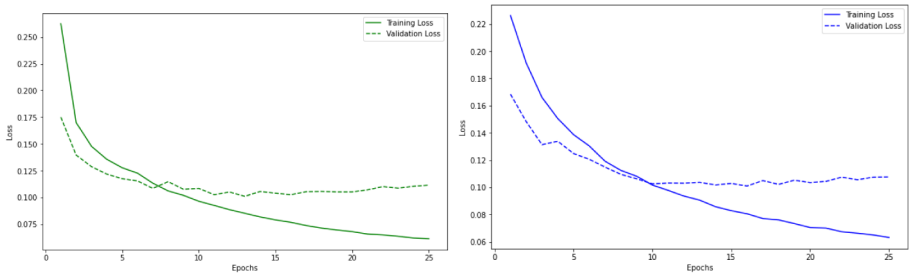
From the results, it is clear that the best performing model employed a combination of bi-LSTM+GRU, along with pre-trained Indian legal BERT embeddings. This approach achieved the results presented in the classification report in Table 14. The classification report shows that the model performed well in structurally classifying legal judgments

Table 12 Classification Report for Indian legal BERT with LDA

Class	Recall	Precision	F1-score	Support
Precedent Analysis	0.86	0.88	0.87	317
Court Discourse	0.66	0.51	0.58	180
Section Analysis	0.74	0.77	0.75	176
Fact	0.76	0.71	0.73	128
Issue	0.84	0.79	0.89	122
Conclusion	0.81	0.94	0.87	114
Argument by Respondent	0.46	0.56	0.50	80
Argument by Appellant	0.47	0.44	0.45	72
Argument by Petitioner	0.26	0.33	0.29	39
Respondent	0.89	0.93	0.91	18
Ratio	0.24	0.19	0.22	12
Appellant	0.65	0.62	0.64	9
Petitioner	0.54	0.65	0.59	8
Judge	0.94	1.00	0.97	6
Argument by Defendant	0.25	0.08	0.12	5
Accuracy			0.72	1286
Macro Avg	0.61	0.64	0.62	1286
Weighted Avg	0.72	0.73	0.72	1286

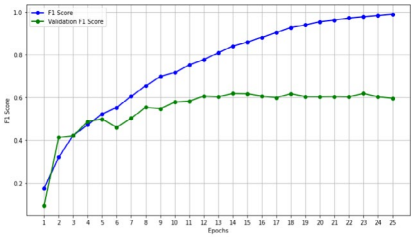
Table 13 Results of DL models with contextual Embeddings

Embedding	Model	Weighted			Macro			Acc
		Recall	Precision	F1-score	Recall	Precision	F1-score	
BERT	LSTM	0.18	0.16	0.15	0.10	0.12	0.11	0.18
	Bi-LSTM	0.69	0.67	0.67	0.55	0.57	0.56	0.69
	Bi-LSTM+GRU	0.69	0.67	0.65	0.62	0.61	0.63	0.7
Distilled Bert	LSTM	0.01	0.02	0.01	0.02	0.01	0.05	0.1
	Bi-LSTM	0.71	0.71	0.71	0.58	0.60	0.58	0.71
	Bi-LSTM+GRU	0.69	0.70	0.68	0.60	0.66	0.60	0.69
Roberta	LSTM	0.20	0.13	0.11	0.09	0.10	0.10	0.20
	Bi-LSTM	0.72	0.69	0.71	0.59	0.60	0.58	0.68
	Bi-LSTM+GRU	0.65	0.65	0.63	0.50	0.60	0.54	0.65
Legal Bert	LSTM	0.27	0.13	0.14	0.08	0.01	0.05	0.27
	Bi-LSTM	0.72	0.71	0.71	0.59	0.62	0.58	0.72
	Bi-LSTM+GRU	0.70	0.72	0.70	0.59	0.62	0.60	0.70
Indian Legal Bert	LSTM	0.27	0.13	0.14	0.08	0.01	0.05	0.27
	Bi-LSTM	0.73	0.72	0.72	0.59	0.62	0.60	0.73
	Bi-LSTM+GRU	**0.75**	**0.76**	**0.75**	**0.70**	**0.68**	**0.69**	**0.75**

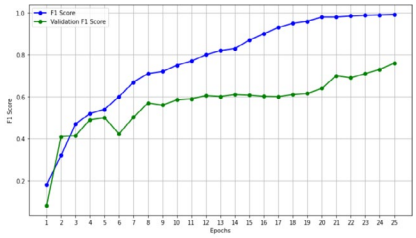


(a) Bi-LSTM with Indian Legal BERT Embedding (b) Bi-LSTM+GRU with Indian Legal BERT Embedding

Fig. 5 Plot of Training Loss and Validation Loss vs Epochs for Indian Legal BERT with DL models



(a) Plot of F1 Score and Validation F1 Score vs Epochs for Indian Legal BERT embedding with Bi-LSTM model



(b) Plot of F1 Score and Val F1 Score vs Epochs for Indian Legal BERT embedding with Bi-LSTM+GRU model

Fig. 6 Comparison of F1 Score and Validation F1 Score for Indian Legal BERT with DL models

Table 14 Classification Report of Indian Legal BERT with Bi-LSTM+GRU

Class	Recall	Precision	F1-score	Support
Precedent Analysis	0.88	0.89	0.89	317
Court Discourse	0.58	0.68	0.62	180
Section Analysis	0.76	0.78	0.77	176
Fact	0.82	0.76	0.79	128
Issue	0.92	0.82	0.87	122
Conclusion	0.90	0.84	0.87	114
Argument by Respondent	0.55	0.56	0.55	80
Argument by Appellant	0.54	0.48	0.51	72
Argument by Petitioner	0.40	0.48	0.44	39
Respondent	0.89	0.89	0.89	18
Ratio	0.23	0.18	0.20	12
Appellant	0.56	0.62	0.59	9
Petitioner	0.88	0.64	0.74	8
Judge	1.00	1.00	1.00	6
Argument by Defendant	0.20	0.50	0.29	5
Accuracy			0.76	1286
Macro Avg	0.67	0.67	0.67	1286
Weighted Avg	0.76	0.76	0.75	1286

into various categories, with an overall accuracy of 0.76. The 'Precedent Analysis' and 'Conclusion' categories had the highest F1-scores of 0.89 and 0.87 respectively, indicating a strong balance between precision and recall. However, the model struggled with the 'Ratio' category, achieving an F1-score of only 0.20. This could be due to the complexity and rarity of legal ratios in the dataset. The 'Argument by Defendant' category also had a low F1-score of 0.29, possibly due to the small number of samples in this class. Also, the 'Argument by Defendant' provides more confusion to the model with respect to other argument terms like 'Argument by Respondent', 'Argument by Appellant' and 'Argument by Petitioner'. Since the number of samples in other arguments are comparatively higher than the 'Argument by Defendant' category, those classes have comparatively higher F1-score (0.44–0.55). However, the F1-score is lesser than non-confusing classes like judge, petitioner, fact, issue etc. Overall, the model's performance suggests that it is effective at structurally classifying legal judgments.

5.5 Fine-tuned BERT based transformer models

Table 15 presents the results of fine-tuned BERT based transformer models, showcasing their performance metrics across various evaluation criteria. Among these models, a merger between legal BERT and Indian legal BERT emerges as the top performer,

Table 15 Results of fine-tuned BERT Transformer Models

MODEL	Weighted			Macro			Acc
	R	P	F1	R	P	F1	
BERT	0.81	0.81	0.81	0.76	0.76	0.76	0.81
Distilled Bert	0.80	0.81	0.80	0.77	0.77	0.77	0.80
Roberta	0.83	0.83	0.83	0.78	0.81	0.79	0.83
Legal Bert	0.83	0.83	0.82	0.77	0.77	0.77	0.83
Indian Legal Bert	0.83	0.84	0.83	0.80	0.81	0.80	0.83
Legal-Indian Legal-SLERP	0.79	0.79	0.78	0.57	0.56	0.56	0.79
Legal-Indian Legal-TIES	**0.84**	**0.85**	**0.84**	**0.80**	**0.82**	**0.80**	**0.84**
Legal-Indian Legal-DARE-TIES	0.82	0.83	0.82	0.75	0.85	0.77	0.83
Legal-Indian Legal-Linear	0.83	0.84	0.83	0.75	0.85	0.77	0.83

Table 16 Classification Report of Legal-Indian Legal BERT-TIES

Class	Recall	Precision	F1-score	Support
Precedent Analysis	0.96	0.92	0.94	317
Court Discourse	0.70	0.74	0.72	180
Issue	0.86	0.83	0.84	176
Argument by Respondent	0.81	0.77	0.79	128
Fact	0.83	0.99	0.91	122
Section Analysis	0.96	0.91	0.94	114
Argument by Appellant	0.85	0.78	0.81	80
Appellant	0.71	0.81	0.75	72
Conclusion	0.67	0.74	0.71	39
Ratio	0.95	1.00	0.97	18
Argument by Petitioner	0.60	0.23	0.33	12
Respondent	1.00	0.90	0.95	9
Petitioner	0.90	1.00	0.95	8
Judge	1.00	0.86	0.92	6
Argument by Defendant	0.50	0.40	0.44	5
Accuracy			0.84	1286
Macro Avg	0.82	0.79	0.80	1286
Weighted Avg	0.85	0.84	0.84	1286

exhibiting notable weighted precision, recall, and F1-score values of 0.85, 0.84, and 0.84 respectively. Further, a combination of fine tuned legal Bert and Indian legal Bert demonstrates consistent and high-quality predictions across different classes, achieving a macro F1-score of 0.80 and an overall accuracy of 0.84. This superiority can be attributed to its specialized fine-tuning on legal datasets, enabling it to grasp the intricacies of legal language effectively. Moreover, Indian legal BERT's enhanced performance underscores its suitability for real-world applications in legal judgment analysis.

Exploration has been done on fine-tuning of various pre-trained BERT based models for structural classification of Indian legal judgments. The Legal-Indian Legal TIES BERT model achieved the best results, with an overall accuracy of 0.84 and a weighted and macro F1 scores of 0.84 and 0.80 respectively, as shown in Table 15. Notably, the Legal-Indian Legal TIES BERT model outperformed the deep learning models, showcasing the efficacy of transfer learning and domain-specific pre-training for structural classification of Indian legal judgments.

The classification report in Table 16 provides a detailed breakdown of the model performance across all classes. Similar trends are observed in Table 16, with 'Argument by petitioner' and 'Argument by Defendant' providing lower F1-scores when compared

to other classes with higher support. This can be attributed to the fact that the contextual embeddings of the words in the vocabulary corresponding to those classes is not representative of the entire data statistics. Additionally, the classes that have overlapping context creates confusion to the model. However, the substantial support count of 1,286 instances ensures the reliability of the reported metrics. These findings underscore the potential of fine-tuning pre-trained language models like Indian Legal BERT/Legal BERT and merging them using various methods such as SLERP and TIES for structural classification of legal judgments in India, paving the way for further advancements in this domain.

6 Discussion

This section further analyzes the results by comparing the best-performing models across various categories, tests the statistical significance of these results by means of statistical testing tools, and further analyzes the class-wise performance of the best models across categories. A post-hoc explainability study is also conducted using tools like SHapley Additive exPlanations (SHAP) [19] and Integrated Gradients (IG) [20].

6.1 Comparison of best performing models

Our research highlights significant distinctions in the performance of ML and DL models within legal judgment analysis. Table 17 presents the performance results of different models across various combinations. Among these models, the merger between legal BERT and Indian legal BERT with TIES optimizer stands out as the best performer across both weighted and macro-average metrics due to using geometrically and semantically informed methods of merging transformer models. Notably, LDA and logistic regression models, when augmented with SMOTE, showcase robustness in handling intricate semantic layers in legal documents. This method, particularly the combination of SMOTE and MLP, stands out, demonstrating the most promise.

Table 17 reports a higher numerical accuracy (0.94) for an MLP model using a specific handcrafted feature engineering and synthetic data generation i.e. n-gram TF-IDF+SMOTE+PCA-200, whereas the Legal-TIES BERT model achieves 0.84 accuracy. The MLP result is obtained using a carefully curated, task-specific feature set that is optimized for this dataset. While this yields strong performance, such handcrafted feature combinations may not generalize well across datasets or legal subdomains. In contrast, Legal-TIES BERT learns contextual representations directly from raw text and demonstrates more consistent performance across settings with the goal of evaluating scalable and domain-adaptive NLP models. Legal-TIES BERT represents an end-to-end approach that does not depend on manually selected linguistic features, making it more

Table 17 Results of best performing models across different combinations

Data Augmentation Strategy	MODEL	Weighted			Macro			Acc
		R	P	F1	R	P	F1	
No data augmentation	n-gram TF-IDF+PCA-500+LDA	0.72	0.73	0.72	0.64	0.69	0.65	0.72
	Indian Legal BERT+LDA	0.72	0.73	0.72	0.61	0.64	0.62	0.72
	Indian Legal BERT+Bi-LSTM+GRU	0.76	0.76	0.75	0.67	0.67	0.67	0.76
	Legal-Indian Legal-TIES	**0.84**	**0.85**	**0.84**	**0.80**	**0.82**	**0.80**	**0.84**
SMOTE-based augmentation	n-gram TF-IDF+SMOTE+PCA-200+MLP	0.94	0.94	0.94	0.94	0.94	0.94	0.94

suitable for real-world deployment where feature extraction pipelines may be costly and often rigid. The MLP model's higher score demonstrates the upper bound achievable with optimized handcrafted features and synthetic data augmentation and dimensionality reduction which is a comparatively long pipeline, and we view this result as complementary rather than contradictory. Observing the results in Table 10, the support value is 317 which represents number of samples in the majority class in the dataset, as evident from Table 6. It highlights that, while classical ML can excel with careful feature design, pretrained transformer models offer a strong and more generalizable alternative with significantly reduced feature-engineering effort.

The integration of SMOTE for data augmentation further enhances its effectiveness in handling imbalanced datasets, leading to superior performance compared to other models with accuracy of 0.94, as provided by Table 10. The scores are better for the classes 'Argument by Defendant' and 'Ratio' which were lower in all other cases without data augmentation. It can be understood that the support values are comparatively higher in SMOTE based method (317) compared to those in other approaches (5 and 12 respectively). Coming back to Table 17, the merger between legal Bert and Indian legal BERT provides a dip in the performance, with F1-score and accuracy of 0.84. However this is more reliable compared to SMOTE as no synthetic data is being generated. Additionally, the combination of Bi-LSTM and GRU with Indian legal BERT also yields competitive results with accuracy 0.76, but it is still slightly lower than the finetuned Indian Legal BERT model. Overall, these findings underscore the effectiveness of fine-tuned Indian legal BERT in legal text classification tasks.

6.2 Statistical significance analysis

To evaluate whether the observed improvements in classification performance are statistically meaningful, we conducted a paired non-parametric significance test using category-wise F1-scores.

Let $C = \{c_1, c_2, \ldots, c_{15}\}$ denote the set of legal discourse categories. For each category $c_i \in C$, we obtain category-wise F1-scores from two models:

- $F1_{A,i}$: F1-score of the proposed Legal-Indian Legal-TIES model,
- $F1_{B,i}$: F1-score of the baseline model.

The paired difference for each category c_i is defined as:

$$d_i = F1_{A,i} - F1_{B,i}. \tag{4}$$

This formulation ensures that each comparison is paired by category, eliminating inter-category variability. Taking into account the problem at hand, the following observations can be made:

- The number of paired observations is small (N=15)
- The distribution of F1-score differences cannot be assumed to be normal
- Only aggregated category-level metrics are available

We employ the paired Wilcoxon signed-rank test [87], a non-parametric alternative to the paired t-test. The Wilcoxon test evaluates whether the median of the paired differences d_i is significantly different from zero. A statistically significant result indicates that the proposed model demonstrates consistent performance improvements across legal

categories, rather than improvements driven by a small subset of classes. The null and alternative hypotheses for the paired Wilcoxon signed-rank test are defined as follows:

$$H_0 : \ \text{median}(d_i) = 0, \tag{5}$$

$$H_1 : \ \text{median}(d_i) \neq 0. \tag{6}$$

We perform a two-sided test at 0.05 significance level. The results of the test are reported in Table 18. For each model comparison, we report the number of categories (N), mean difference in F1-scores ($\triangle$F1), median difference in F1-scores, Wilcoxon test statistic (W), p-value and statistical significance at $\alpha = 0.05$. A result is considered statistically significant if $p < 0.05$. α is set to 0.05 because it is the standard threshold that limits false positives to 5% and is widely accepted in machine learning evaluation.

Table 18 reports paired Wilcoxon signed-rank test results comparing the proposed best model viz. Legal-Indian Legal-TIES with baseline approaches mentioned in Table 17 across 15 legal categories. The proposed model achieves consistent improvements, with mean $\triangle$F1 values of 0.1300, 0.1727, and 0.1493, and corresponding median $\triangle$F1 values of 0.1200, 0.1400, and 0.1300 when compared with Indian Legal BERT + BiLSTM-GRU, Indian Legal BERT + LDA, and TF-IDF + PCA + LDA, respectively. All comparisons yield statistically significant results with p-values of 0.0248, 0.0084, and 0.0106, which are below the significance threshold $(\alpha = 0.05)$, confirming that the observed F1-score gains are consistent across legal categories rather than driven by isolated classes.

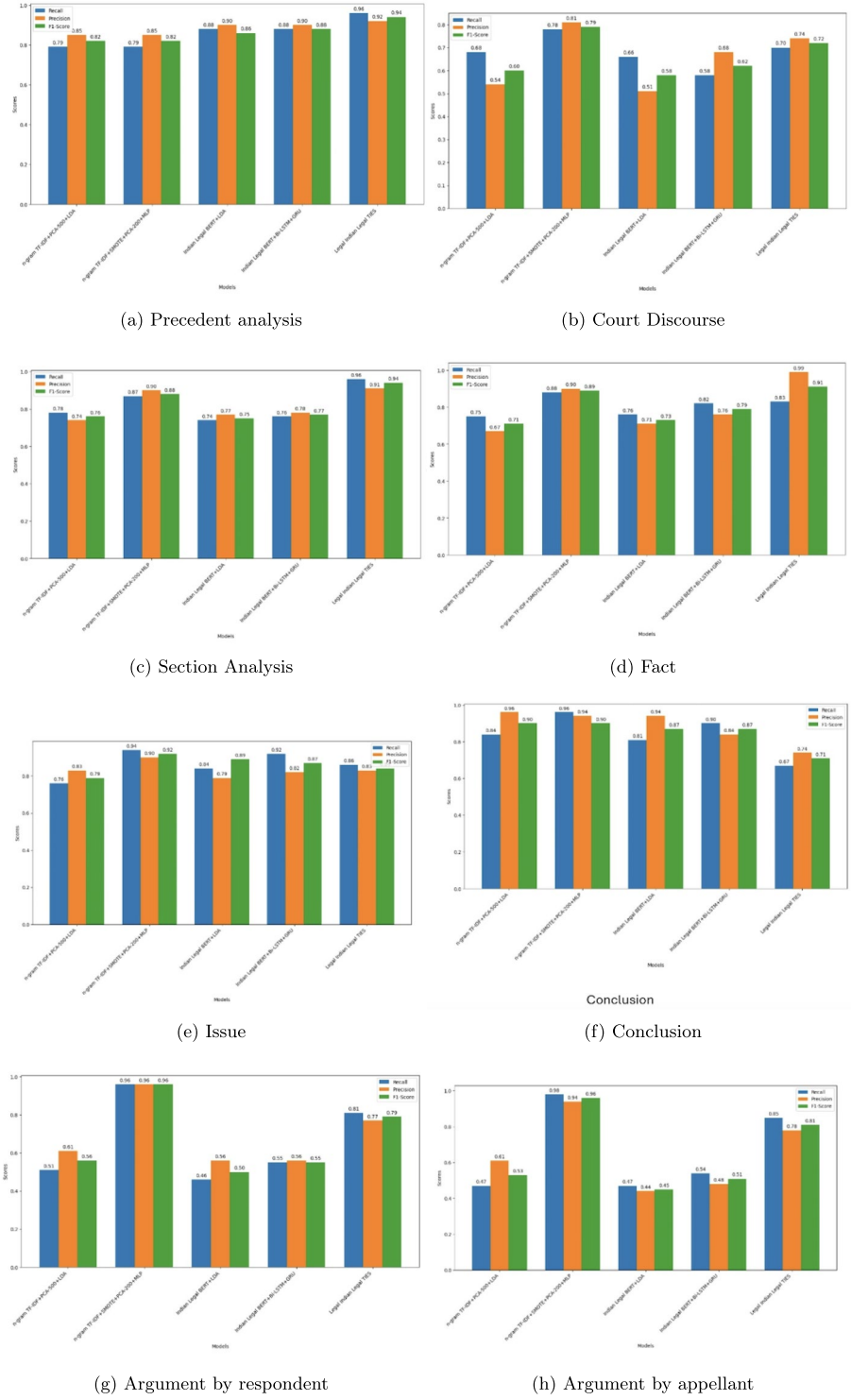
6.3 Class-wise performance analysis of best models

Fig. 7 highlights the class-wise performance for structural classification of Indian legal judgments of the five best performing models from our research. Recall is consistently high for all models, indicating that they are good at retrieving relevant sections. The 'judge' class gives the best recall, precision, F1-score. Judge category texts are usually short thereby facilitating more efficient classification. Respondent class also gives relatively high score compared to other classes. 'Argument by Defendant' class shows poor performance in all models except TF-IDF+SMOTE+PCA+MLP. Since this class incorporates the issues presented by the defendant, they varied and cannot be predicted to a great extent.

In terms of F1 score, TF-IDF+SMOTE+PCA+MLP model seems to show slightly better performance. Recall and F1 Scores are consistently high across the board, but for 'Fact' class, the model shows significantly better performance in F1 Score for the above model. For 'Fact' and 'Issue', recall is generally higher, meaning that most models are good at retrieving relevant data, but precision may need improvement (indicating some false positives). For 'Argument by Respondent' class, the trend is slightly different. Here,

Table 18 Paired Wilcoxon Signed-Rank Test Comparing Proposed Legal-Indian Legal-TIES with Baseline Models on Category-wise F1-Scores

Model	N (Categories)	Mean Δ F1	Median ΔF1	Wilcoxon *W*	*p*-value	Significant $\alpha (= 0.05$
Indian Legal BERT + BiLSTM-GRU	15	0.1300	0.1200	20.50	0.024839	✓
Indian Legal BERT + LDA	15	0.1727	0.1400	15.00	0.008362	✓
TF-IDF + PCA + LDA	15	0.1493	0.1300	15.00	0.010578	✓



(a) Precedent analysis (b) Court Discourse

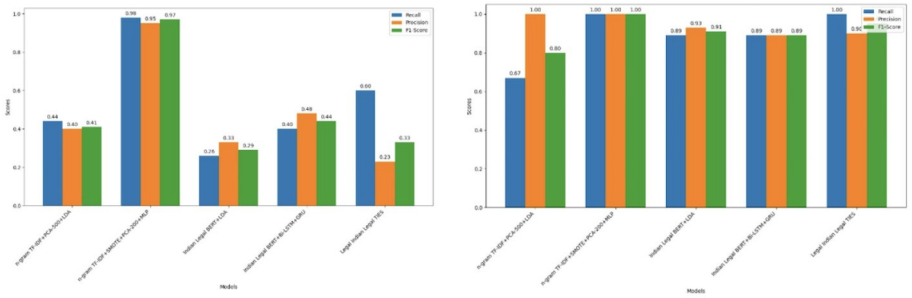
(c) Section Analysis (d) Fact

(e) Issue (f) Conclusion

(g) Argument by respondent (h) Argument by appellant

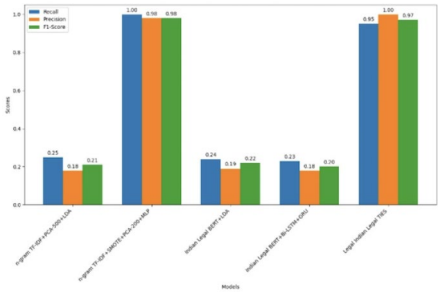
Fig. 7 Comparison of performance of the best performing model across various classes

the bi-LSTM + GRU combination is observed to give better scores across all metrics considered. In categories like 'Argument by Respondent' and 'Appellant', more variation is observed across models, suggesting that some tasks may be more challenging to predict or classify. Across most categories, the TF-IDF+SMOTE+PCA+MLP model seems to perform consistently well in terms of F1 score. SMOTE with MLP is giving the best

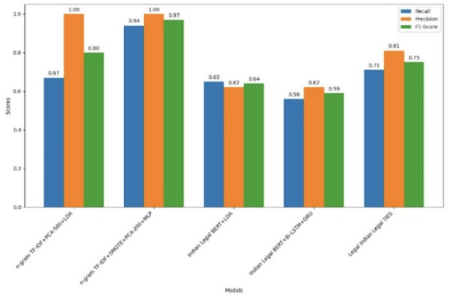


(i) Argument by petitioner

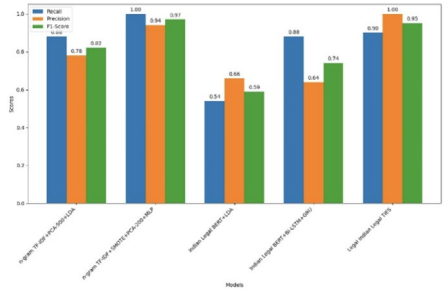
(j) Respondent



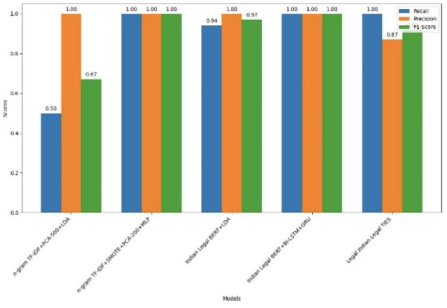
(k) Ratio



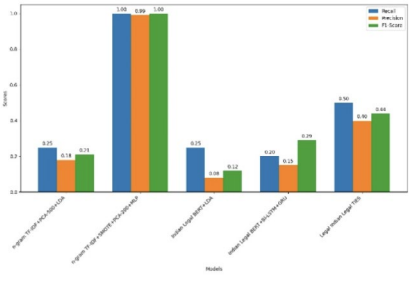
(l) Appellant



(m) Petitioner



(n) Judge



(o) Argument by defendant

Fig. 7 (continued)

results for most classes, as the synthetically generated data is used to balance the number of class samples.

The findings from our study have important implications for the field of legal analytics and practice. Using AI to accurately and efficiently analyze legal texts can make processes like document review, case prediction, and legal research more effective. This could make the legal system work more smoothly and reduce the workload on legal professionals, like lawyers and paralegals. Additionally, providing better tools for

legal analysis through AI could make legal resources more accessible to everyone. This research is crucial for professionals working in AI in the legal domain, as it offers practical insights and tools that can significantly improve their work.

6.4 Post-hoc explainability study

In multi-class text classification, it is important not only to predict class labels but also to explain why a document is not associated with other plausible classes or positioned near a decision boundary. This perspective is particularly critical in legal and judiciary where subtle distinctions between categories can have significant consequences. To support this deeper level of analysis, our framework enables class-wise explainable AI (XAI), allowing explanations to be generated for any chosen class. In this research involving structural judgement classification, the best-performing legal-Indian legal BERT-TIES model is subjected to explainability analysis using the SHAP [19] and IG [20] tools. In this work, SHAP is used for estimating token-level contribution to model predictions, thereby providing a post-hoc, game-theoretic explanation for the influence of legal terms on classification outcomes. Class-wise explainability enables explanations to be generated for any target class, providing a deeper understanding of how input features influence each class independently. The results for class-wise explainability using a combined SHAP and IG strategy are used, and heatmaps are generated. Even though experiments have been conducted for all classes, we use the classes as provided in Fig. 8 for visualization. This allows us to understand which parts of the text contribute positively or negatively to the model likelihood of classifying the text as a particular structural class.

The Court Discourse heatmap highlights procedural and judicial terminologies like 'convened' and 'issuance'. These words receive positive attributions due to their influence on court procedural activity. In contrast, statutory references and constitutional framing exhibit weaker influence. In section analysis, the clauses including 'companies', 'petitioner' etc are given higher importance. In fact class, words relating to contextual background needed to identify factual sections like 'apply' are highlighted as primary tokens. Issue class is related to legal disputes and questions and clauses like 'convened',

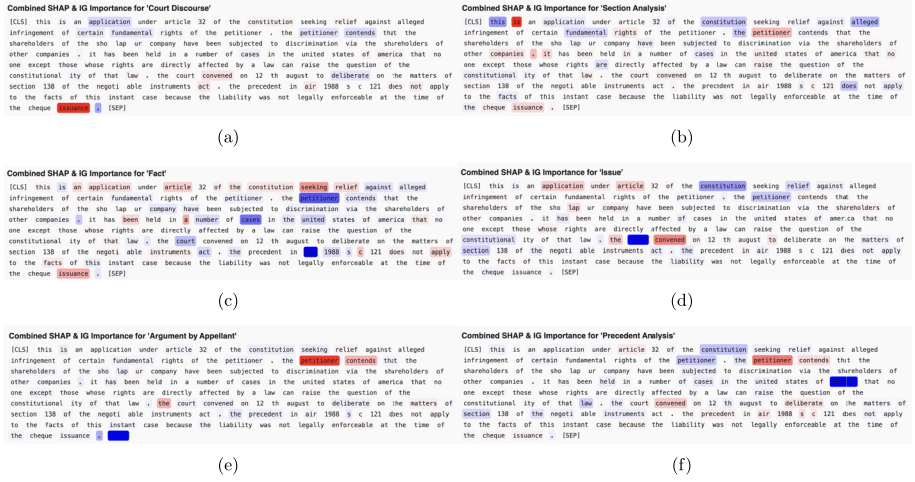


Fig. 8 Combined SHAP & IG Importance identifying the top tokens (red highlighted) for **a** Court Discourse, **b** Section Analysis, **c** Fact, **d** Issue, **e** Argument by Appellant, **f** Precedent Analysis for a sample text from Indian legal judgements. Each heatmap highlights token-level contributions computed with respect to the corresponding class, enabling a direct comparison of how the same input text is interpreted differently across classes

'application', 'article' etc are given primary attributions. Argument by Appellant class correctly gives fair importance to terms like 'petitioner' and 'contends', both representative of the intent of argument. Similarly, 'petitioner', 'contends' and 'convened' are recognised as the top tokens in precedent analysis. The experiments have been conducted for all 15 classes. However, for visual representation, six classes have been selected from the overall class structure and represented in Fig. 9. Collectively, these contrasting and sometimes overlpping attribution patterns confirm that the model learns legally meaningful representations from the proposed class-wise XAI framework and provides faithful insights into how various legal discourse roles can be analyzed from the same segment of text.

Apart from getting the heatmap-based attributions for each class, the prediction model for a sample text is analyzed using SHAP, IG and attention-based explainability suite. We propose a comprehensive explainability approach that combines multiple interpretability tools to analyze transformer-based structural judgment classification at the token level. Model attention weights are used to highlight tokens that receive greater contextual focus, while gradient-based analysis of the target class with respect to input embeddings captures sensitivity of prediction to individual tokens. In addition, token contributions are estimated using an approximation of integrated gradients.

In order to illustrate the complementary perspectives in interpreting token importance, the plots in Fig. 9 are generated using a sample text and the legal-Indian legal BERT ties model. Figure 9 represents the SHAP, attention and IG plots corresponding to the importance of tokens in the sample text. From Fig. 9(a), it can be observed that attention plot shows relatively uniform distribution among the top tokens. In contrast, Fig. 9(b) demonstrates that SHAP plot provides a steeper distinction among the top tokens. The tokens like 'this', 'petitioner', 'contends', 'constitution' are receiving higher prominence. The IG plot shows a smoother decay than SHAP, emphasizing tokens that make consistent contributions to the prediction while suppressing sensitivity effects due to noise. Notably, legally meaningful terms appear prominently in both SHAP and IG, while attention alone is less discriminative. While SHAP provides a sharper decay in scores, Fig. 9(c) shows that IG shows smoother decay which emphasises that the tokens 'petitioner', 'convened' and 'contends' show higher attribution while not taking into account the stop words that appear to have higher contribution. Based on these

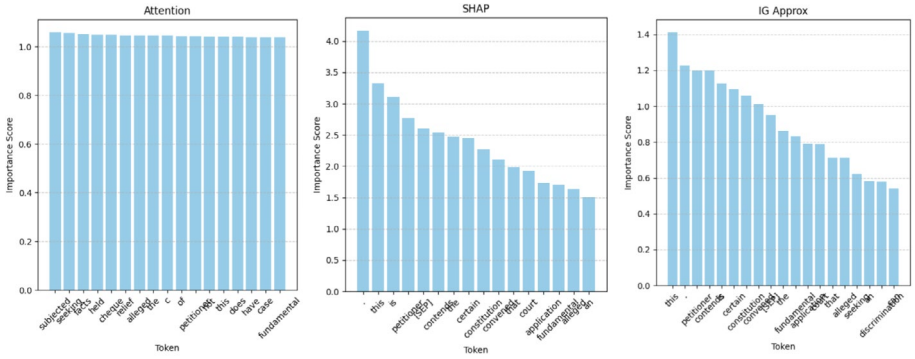


Fig. 9 The summary plots show the token importance in the sample text evaluated using three different explainability techniques **a** Attention, **b** SHAP, and **c** Integrated Gradient for the structural judgement classes. The prediction for this particular text segment was derived as 'Precedent Analysis' from the legal-Indian legal TIES model

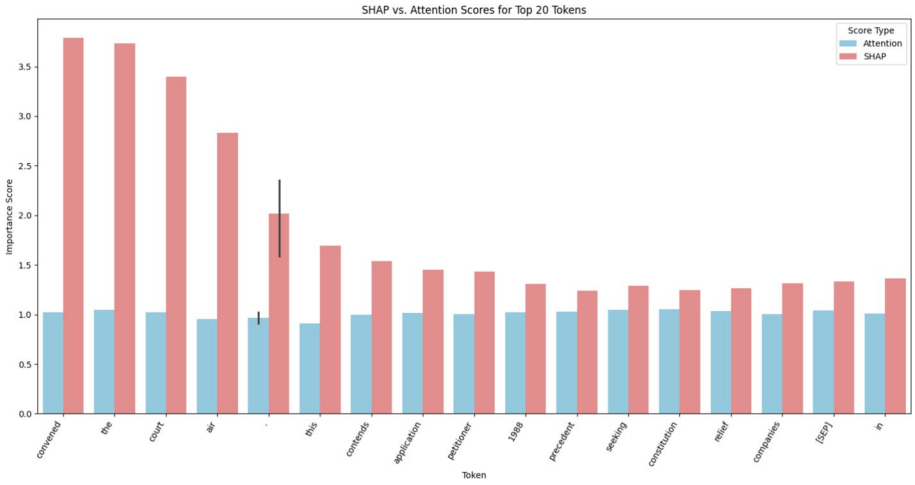


Fig. 10 Comparison of SHAP versus Attention importance scores for the top 20 tokens from a sample judgement text belonging to 'Precedent Analysis' class

attributions, the model gave the predicted class for this sample text as 'precedent analysis', consistent with the tokens highlighted in Fig 8(f).

Further, a comparison of SHAP versus attention plot for the top 20 tokens in a sample judicial text is analyzed in Fig. 10. Each token is represented by two bars: the blue bars indicate attention weights, while the red bars represent SHAP based importance. From Fig. 10, it can be observed that attention scores remain relatively uniform across tokens. In contrast, SHAP assigns substantially higher importance to particular tokens, viz., 'convened,' 'court,' 'AIR,' and 'petitioner.' These tokens play a prominent role in the model's decision-making. There is a clear mismatch between the high SHAP scores and predominantly flat attention score for the document. It provides the understanding that high attention does not necessarily mean that those tokens predominantly influence the model's decisions. Attention mainly depicts where the model is focusing as it processes the text, whereas SHAP highlights which words actually influence the final prediction. As a result, common function words or routine legal connectors may attract attention without playing a meaningful role in the decision, while legally significant terms stand out in terms of higher importance even when their attention scores are comparable. By merging these complementary perspectives, the framework derives a unified importance score that offers a more stable and meaningful interpretation of model behavior.

7 Conclusions and future scope

This study examines how different learning-based models perform when applied to the analysis of legal judgments, with particular attention to the role of representations and feature-handling techniques such as embeddings, SMOTE, and PCA for dimensionality reduction. The work adopts a broad experimental perspective, covering a range of methods commonly used in the legal judgment analysis domain. These include traditional machine learning models, such as support vector machines and logistic regression, as well as more recent deep learning approaches, including BiLSTM GRU architectures and transformer based Legal BERT embeddings. In addition, multiple variants of LegalBERT are compared and combined using different merging strategies to assess their complementary strengths. Among the feature-based approaches explored, the n-gram TF-IDF + SMOTE + PCA-200 + MLP configuration achieved the highest observed accuracy

of 0.94. The results suggest that effective structural classification of legal judgments does not necessarily require manual feature engineering or data augmentation. While feature-engineered models reach higher peak accuracy, the combined LegalBERT and Indian LegalBERT model provides more stable and balanced performance when operating directly on raw text, attaining a weighted F1-score of 0.84 and a macro F1-score of 0.80. Overall, these findings point to the practical value of domain-adapted transformer models for scalable legal analytics, particularly in settings where handcrafted features or synthetic data generation are undesirable.

Future research can build on this work by exploring more fine-grained structural categories and adopting advanced deep learning models, including legal-specific large language models. Extending the approach to cross-lingual and cross-jurisdictional settings would improve its applicability across different legal systems. Incorporating legal precedents with socio-economic and other external data sources may also enable more context-aware analysis. Finally, developing standardized benchmarks for structural legal judgment classification would support fair comparison and evaluation of future legal analytics systems.

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Author Contributions
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Data Availability
The datasets generated during and/or analysed during the current study are available from the corresponding author on reasonable request.

Declarations

Ethics approval
Not applicable.

Consent to participate
Not applicable.

Consent for publication
Not applicable.

Conflict of interest
The authors declare no Conflict of interest.

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